

Finite reservoirs at phase coexistence: full-state accuracy and phase correlations

Abstract

How large must a thermal reservoir be for a subsystem in contact with it to have its canonical distribution? We couple a subsystem of size N weakly to a reservoir with density of states proportional to U^{c_N} (heat capacity $c_N k_B$), choose the total energy as favorably as possible, and measure the total variation (TV) distance between the subsystem's complete microscopic distribution and the canonical one. For a single phase with ordinary $\sqrt{N}$ energy fluctuations, $c_N \gg N$ is necessary and sufficient, as expected from equivalence of ensembles. At a first-order transition the answer depends on the number of distinct phase energies. With three or more, no choice of total energy helps and $c_N \gg N^2$ is required. With two, as at a temperature-driven transition, choosing the total energy so that the reservoir weights both phase energies equally reduces the requirement to $c_N \gg \Delta_N s_N$, where Δ_N is the latent-energy gap and s_N the nondegenerate within-phase fluctuation scale; typically $c_N \gg N^{3/2}$. This is far below the subsystem's own canonical heat capacity at coexistence, which is of order N^2 because of the latent heat. Sufficiency also needs tail bounds within each phase and a bound on configurations in neither phase, such as interfaces. We prove both directions for the mean-field three-state Potts model and for nearest-neighbor Potts models with many colors in every dimension $d \geq 2$. At $c_N \sim \gamma N^{3/2}$, under stronger phase-tail and exceptional-mass assumptions (proved for $d = 2$ only for large γ), we compute the limiting error: adjusting the total energy restores the phase weights but not the energy distribution within the phases, and the smallest error may come from suppressing one phase. Two independent copies sharing a reservoir with $N \ll c_N \ll N^2$ can be driven into opposite phases. Their mutual information then tends to one bit, and with balanced phase weights each copy is asymptotically canonical while the joint TV error tends to $1/2$; joint accuracy requires $c_N \gg N^2$. Earlier work described how finite reservoirs distort coexistence histograms and gave sufficient sizes of order N^2 through a general thermalization bound; we give two-sided thresholds for the complete microscopic law and prove them for interacting Potts models.

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1 Introduction

A subsystem in weak contact with a finite reservoir does not have exactly its canonical distribution, because the reservoir entropy is a concave rather than linear function of energy. Away from phase transitions this matters little: nearly all canonical probability of a subsystem of size N lies in an energy window of width of order $\sqrt{N}$, and a reservoir whose heat capacity grows faster than N reproduces the canonical law. At first-order coexistence the canonical law has two or more such windows, separated by energies of order N . A reservoir that is accurate inside one window can then change the relative weights of the phases, the fluctuations inside them, or both. This paper determines how large a reservoir must be to reproduce the complete canonical law of an interacting subsystem at coexistence, when the total energy of the isolated composite is chosen as favorably as possible.

The question arises whenever a system exchanges energy with an environment not much larger than itself. Finite-bath and Gaussian-ensemble Monte Carlo methods use such reservoirs deliberately to sample first-order transitions [6, 20], and a subsystem of a larger closed system thermalizes only as accurately as the remainder allows [30]. In such settings the reservoir heat capacity is expected to control the error, and the practical question is how fast it must grow with subsystem size at coexistence.

Setting. The subsystem has energy E_N and canonical law P_N at a fixed inverse temperature. The reservoir has density of states $\omega_B(U) \propto U^{c_N}$ for $U > 0$, the form produced by $2(c_N + 1)$ classical quadratic degrees of freedom. We call c_N the reservoir capacity: it is the reservoir heat capacity in units of k_B (Section 2 fixes the entropy convention). The coupling is weak, and the only adjustable quantity is the total energy $\mathcal{E}_N$ of the isolated composite; we call its choice the calibration. We measure accuracy by the total variation distance $d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)|$ between the subsystem's reservoir law Q_N and P_N on the full microscopic state space, and call it the full-state TV. When it is small, every bounded observable, including the phase label and the energy fluctuations within a phase, has nearly its canonical distribution; matching thermodynamic potentials or a few averages does not guarantee this. Because the reservoir reweights states only through their energy, the full-state TV equals the TV between the two energy distributions (Lemma 2.1). It can therefore be computed from energy laws without replacing the microscopic system by an energy-only model. We write $x_N \gg y_N$ when $x_N/y_N \rightarrow \infty$.

1.1 The mechanism

Up to a constant, the reservoir multiplies the canonical weight of a subsystem state of energy e by $\exp[\beta e + c_N \log(\mathcal{E}_N - e)]$. The exponent is a concave function of e with curvature of order $1/c_N$ at the relevant energies. Suppose first that three energies spread over a common scale b_N must all keep their canonical relative weights. A concave function with curvature of order $1/c_N$ cannot take nearly equal values at three such points unless $b_N^2/c_N \rightarrow 0$, and no choice of $\mathcal{E}_N$ changes this. Two phases behave differently. Choosing $\mathcal{E}_N$ so that the exponent takes equal values at two phase centers separated by Δ_N , which we call the secant calibration, leaves slopes of order Δ_N/c_N at the centers. Across a phase of energy width s_N , such a slope changes the log weight by order $\Delta_N s_N/c_N$, so the phase is preserved only if $c_N \gg \Delta_N s_N$. For an extensive latent heat and $\sqrt{N}$ fluctuations this is $c_N \gg N^{3/2}$, well below the N^2 required for three distinct phase energies.

Table 1: Reservoir capacities needed for full-state TV convergence, optimized over the calibration. The entries are asymptotic theorems under the stated hypotheses, not finite-size error bounds.

Canonical energy structure	Required capacity	Hypotheses and result
One ordinary fluctuation window	$c_N \gg N$	Nondegenerate weak Gaussian limit on scale $\sqrt{N}$; Theorem 3.1
Two extensive energy phases	$c_N \gg N^{3/2}$	One nondegenerate $\sqrt{N}$ limit, exponential phase tails, and small exceptional mass; Corollary 3.6
At least three extensive energy phases	$c_N \gg N^2$	Weak macroscopic limit with at least three support points; Theorem 3.1
Two independent two-phase copies, joint target	$c_N \gg N^2$	Combined energy of the copies has three support points; Corollary 5.3

1.2 Main results

We prove two-sided capacity thresholds for the complete state law, verify their hypotheses for interacting Potts Hamiltonians, compute the exact error at the threshold scale and optimize it over all calibrations, and determine the marginal, joint, and information limits for two systems sharing a reservoir.

Sharp criteria (Section 3). Suppose that on some scale b_N the energy concentrates on at least three distinct values, in the sense that $(E_N - a_N)/b_N$ converges weakly to a law with at least three support points. Then some calibration makes the full-state TV vanish if and only if $c_N \gg b_N^2$ (Theorem 3.1); no density or moment bound is needed. A single phase with a nondegenerate Gaussian limit on the $\sqrt{N}$ scale thus requires $c_N \gg N$, and three or more distinct extensive phase energies require $c_N \gg N^2$. For two phases with gap Δ_N and fluctuation scale s_N , the condition $c_N \gg \Delta_N s_N$ is necessary for every calibration as soon as one phase has a nondegenerate fluctuation limit (Theorem 3.4). It is sufficient, with the secant calibration, under exponential moment bounds within each phase and a bound on the exceptional mass, the canonical probability of configurations assigned to neither phase, such as those with macroscopic interfaces (Theorem 3.5 and Corollary 3.6). What matters is the number of distinct phase *energies*: symmetry-related ordered phases with the same energy do not create a three-energy obstruction. Table 1 summarizes the thresholds.

Microscopic models (Section 4 and Appendix A). The threshold is exactly $c_N \gg N^{3/2}$ for the three-state Curie–Weiss–Potts model and for the nearest-neighbor Potts model with a fixed, sufficiently large number of colors in every dimension $d \geq 2$ (Theorems 4.1 and 4.3), with or without independent kinetic energy (for the short-range model, see the end of Appendix A). For the short-range model the difficulty is sufficiency: the secant calibration can raise the weight of rare interfacial configurations by up to $\exp[O(N^2/c_N)]$. We therefore need a bound on the canonical probability of these configurations themselves; a signed error bound on the partition function alone does not give this control. We obtain it from positive restricted contour partition functions, which also give exponential moments of the phase energies, and we transfer these moments from the number of occupied bonds to the spin energy through the Edwards–Sokal coupling.

A reservoir smaller than the subsystem’s heat capacity. In both models, with limiting phase weights w_-, w_+ (the canonical probabilities of the two phases), the subsystem’s own canonical

heat capacity is of order N^2 at coexistence:

$$\frac{C_{S,\text{can}}}{k_B N^2} = \frac{\beta_N^2 \text{Var}_{P_N}(E_N)}{N^2} \longrightarrow \beta^2 w_- w_+ \ell^2, \quad \ell = \lim_N \Delta_N / N. \quad (1.1)$$

Indeed, the bounded spin energy per particle converges in law to a two-atom law, and optional independent kinetic energy with Gamma shape aN ($a = 0$ meaning none) adds only aN/β_N^2 to the variance. The N^2 term comes from the latent heat; the within-phase variances are at most of order N . Yet a reservoir with $N^{3/2} \ll c_N \ll N^2$, which is still much larger than the subsystem, reproduces the coexisting law. Absorbing the latent heat changes the reservoir temperature enough to bias the phase odds by a factor of order $\exp(N^2/c_N)$, but the secant calibration cancels this bias. What remains is a change of order $N^{3/2}/c_N$ in the log weight across each phase, and this sets the threshold.

The threshold regime (Section 6). At the boundary scale $c_N \sim \gamma N^{3/2}$, the energy distribution of each phase is exponentially tilted, which shifts it by a finite number of standard deviations. We compute the limiting full-state TV exactly (Theorem 6.2). A further calibration shift of order $\sqrt{N}$ restores the phase weights but not the within-phase distributions of phases with nondegenerate fluctuations (Corollary 6.3). Optimizing over every calibration gives the smallest achievable limiting error (Theorem 6.4), which can be attained by discarding one phase entirely; an optimized distance should therefore be reported together with the resulting phase weights. These results use stronger phase-tail and exceptional-mass hypotheses, which hold for every $\gamma > 0$ in the mean-field model and in the short-range model for $d \geq 3$, and in two dimensions for sufficiently large γ .

Shared reservoirs (Section 5 and Appendix D). Consider two otherwise independent copies of a two-phase system coupled to one reservoir with $N \ll c_N \ll N^2$, and place the reservoir maximum at the opposite-phase pair energy $e_{-,N} + e_{+,N}$, midway between the two same-phase energies. Same-phase pairs are then suppressed, while the ordinary fluctuations inside each phase survive (Theorem 5.1). The mutual information between the two full microscopic states tends to one bit for any phase weights (Proposition 5.2). With balanced phase weights, each copy's full-state law also converges to its canonical law, whereas the joint TV error tends to $1/2$. In the spin models, balance is obtained by shifting the reference temperature by order $1/N$ from coexistence; the reservoir uses the same shifted temperature, and the target is the balanced canonical law at that temperature. Joint canonical accuracy requires $c_N \gg N^2$, because the combined energy of two two-phase copies has three macroscopic support points (Corollary 5.3). Appendix D treats a prescribed number of copies in each phase among m copies, and the correlated fluctuations that appear when c_N is proportional to N .

Exactly solvable models and diagnostics (Appendices B and C). For Gaussian phases that exchange several quantities with a reservoir of isotropic quadratic entropy, the threshold is on the $N^{3/2}$ scale if the phase points lie on a common sphere and on the N^2 scale otherwise (Theorem B.2); between these scales we compute the smallest achievable error, which is the weight of the phases that must be discarded (Theorem B.3). The reservoir can raise the weight of an interfacial configuration by at most $\exp[O(N^2/c_N)]$, while its canonical weight is suppressed by a surface cost of order $N^{(d-1)/d}$. These balance at $c_N \asymp N^{1+1/d}$, which is below $N^{3/2}$ for $d \geq 3$ and equal to it for $d = 2$; a stated two-dimensional capillarity model makes this competition precise and motivates the large- γ restriction above. Finally, two examples show that an unweighted Euclidean histogram norm can vanish while the full-state TV does not, and that, for an extensive gap under the secant calibration,

absolute accuracy of the log-density barrier between a phase center and the interphase midpoint requires $c_N \gg N^2$.

Calibration precision. The secant calibration must fix the reservoir energy, or equivalently its temperature at the phase centers, to relative accuracy $o(1/N)$ to preserve the phase weights; a relative error of order $1/N$ changes the phase odds by a factor of order one. This is no stricter than the canonical target itself, whose phase odds change by a factor of order one when β changes by order $1/N$. Section 6 quantifies calibration errors of order $\sqrt{N}$ in $\mathcal{E}_N$ at the threshold scale.

1.3 Relation to earlier work

Finite-reservoir ensembles and their fluctuation corrections are well established. Challa and Hetherington introduced Gaussian ensembles as a controllable interpolation between familiar ensembles and showed how they change first-order energy histograms [7, 6]. A Gaussian ensemble is the quadratic approximation of the reservoir used here, with curvature about β^2/c_N . The two-Gaussian coexistence description of Challa, Landau, and Binder [8] already separates an extensive latent energy gap from ordinary within-phase widths, and their later review covers finite-size and finite-bath approaches [9]. Within such a two-Gaussian picture, the mechanism above gives the $N^{3/2}$ scale heuristically; our contribution is the two-sided theorem, its precise tail conditions, and its proof for interacting Hamiltonians. Power-law subsystem laws from finite reservoirs of constant heat capacity are standard [5], and the surface and volume entropy conventions for the reservoir are compared in [21, 17].

For a single phase, the condition $c_N \gg N$ is the expected equivalence-of-ensembles scale. It parallels the sharp growing-block conditional limit theorems of Diaconis and Freedman [16], which treat a regular exponential-family conditioning problem rather than the phase mixture of an interacting subsystem coupled to a separate reservoir. Riera, Gogolin, and Eisert [30] control a reduced microcanonical-shell state in trace distance by the nonlinear remainder of the reservoir entropy. In their reservoir example a sufficient size is proportional to the square of the subsystem energy range; in a commuting setting this is a TV bound of the N^2 type. A bound based on the full energy range is not sharp for a single phase, where $c_N \gg N$ suffices. We show that two coexisting phases need only $N^{3/2}$ after calibration, and that N^2 is sharp when three distinct phase energies must be preserved. Our results concern classical systems, described by probability measures on the state space.

The closest numerical precedent is the finite-reservoir Potts study of Griffin, Matty, and Swendsen [20]. They measure agreement by an unweighted Euclidean norm of energy-histogram differences and optimize a comparison temperature; Appendix C gives an exact example in which that norm vanishes while the full-state TV does not. Generalized Gaussian ensembles recover nonconcave entropies and macrostate equivalence at a fixed penalty [13, 14], and squeezed ensembles select first-order states with finite-size conversion formulas [31]. Both address thermodynamic-limit questions at a fixed penalty, whereas we study the complete law as the reservoir curvature shrinks with N .

For shared reservoirs, the ensemble dependence of fluctuations and correlations is classical [25, 27]. Ramírez-Hernández, Larralde, and Leyvraz found opposite-phase assignments in two weakly coupled identical systems with negative specific heat [29], and Cohen, Rittenberg, and Sadhu studied ensemble-dependent shared information, including order-one contributions from degenerate phases [10]. We add the capacity window in which a separate reservoir selects opposite phases, and we prove the limits of the full marginal error, the joint error, and the microscopic mutual information in that window.

1.4 Organization

Section 2 derives the exact reservoir law and the centered and secant calibrations, and Sections 3–6 contain the results summarized above. Section 7 presents exact finite-size calculations for the mean-field model, and Section 8 discusses scope and open questions.

2 Reservoir model and accuracy criterion

2.1 Canonical reference and the exact composite marginal

Let $(\Omega_N, \mathcal{F}_N, \mu_N)$ describe the subsystem states, with measurable finite energy $E_N : \Omega_N \rightarrow \mathbb{R}$; μ_N may be counting or classical phase-space measure. At the specified inverse temperature $\beta_N > 0$, assume

$$P_N(\mathrm{d}x) = \frac{e^{-\beta_N E_N(x)}}{\mathcal{Z}_N} \mu_N(\mathrm{d}x), \quad 0 < \mathcal{Z}_N < \infty. \quad (2.1)$$

Throughout, $\beta_N \rightarrow \beta \in (0, \infty)$; letting β_N vary allows a target at finite-size coexistence with equal phase weights. The target law is fixed before calibration.

We neglect the subsystem–reservoir interaction energy and take the reservoir to have the exact surface density of states

$$\omega_{B,N}(U) = A_N U^{c_N} \mathbf{1}_{\{U>0\}}, \quad A_N > 0, \quad c_N > 0. \quad (2.2)$$

(“Surface” means the density at an energy, not its cumulative integral.) Integrating over the composite’s microcanonical shell at total energy $\mathcal{E}_N$ gives the subsystem law

$$Q_N(\mathrm{d}x) = \frac{L_N(E_N(x))}{Z_N} P_N(\mathrm{d}x), \quad L_N(e) = \begin{cases} e^{\beta_N e} (\mathcal{E}_N - e)^{c_N}, & e < \mathcal{E}_N, \\ 0, & e \geq \mathcal{E}_N, \end{cases} \quad Z_N = \mathbb{E}_{P_N} L_N(E_N). \quad (2.3)$$

No quadratic expansion of the reservoir entropy is involved. The exponent c_N is prescribed; only $\mathcal{E}_N$ can be chosen (the calibration). For finite $\mathcal{E}_N$, L_N is bounded: it is continuous below the cutoff and tends to zero at the cutoff and as $e \rightarrow -\infty$. Thus $Z_N < \infty$, and $\mathcal{E}_N$ is admissible when $P_N(E_N < \mathcal{E}_N) > 0$, which is exactly $Z_N > 0$.

With $S_{B,N}(U) = k_B \log \omega_{B,N}(U)$ (a fixed reference energy absorbed in the additive constant), the reservoir temperature T_B satisfies $1/T_B = \partial_U S_{B,N} = k_B c_N / U$. Hence

$$\beta_B = \frac{1}{k_B T_B} = \frac{c_N}{U}, \quad C_B = \frac{\mathrm{d}U}{\mathrm{d}T_B} = k_B c_N, \quad (2.4)$$

so the reservoir capacity c_N is the surface-convention (microcanonical) heat capacity in units of k_B . At canonical inverse temperature β_{can} , the same reservoir has energy density proportional to $U^{c_N} e^{-\beta_{\text{can}} U} \mathbf{1}_{\{U>0\}}$, with mean energy $(c_N + 1)/\beta_{\text{can}}$ and canonical heat capacity $k_B(c_N + 1)$. This offset does not affect the diverging scales proved below. The theorems concern the specified density of states and need not choose between the surface and volume entropies compared in [21, 17].

2.2 Full-state and energy-law accuracy coincide

For probability measures P, Q on the same space,

$$d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2} \int |\mathrm{d}P - \mathrm{d}Q|. \quad (2.5)$$

The last integral uses any common dominating measure. Let P_N^E, Q_N^E be the pushforwards by E_N .

Lemma 2.1 (Energy is sufficient for reservoir reweighting). *For the exact marginal (2.3),*

$$d_{\text{TV}}(Q_N, P_N) = d_{\text{TV}}(Q_N^E, P_N^E) = \frac{1}{2} \mathbb{E}_{P_N} \left| \frac{L_N(E_N)}{Z_N} - 1 \right|. \quad (2.6)$$

Proof. The Radon–Nikodym derivative on the state space is $L_N(E_N)/Z_N$. Its pushforward derivative is the same function of energy: for every bounded measurable f ,

$$\mathbb{E}_{Q_N} f(E_N) = \frac{1}{Z_N} \int f(e) L_N(e) P_N^E(\text{d}e).$$

The two L^1 formulas therefore agree. $\square$

Matching a mean energy, a pressure, or a few local observables need not determine the full-state TV. Total variation controls uniformly bounded observables; unbounded observables also need tail control, such as uniform integrability under both laws. Lemma 2.1 relies on the energy-only likelihood and can fail when the coupling changes the subsystem Hamiltonian.

Lemma 2.2 (Normalization criterion). *Suppose $W_N \geq 0$ and $\mathbb{E}_{P_N} |W_N - 1| \rightarrow 0$. Then $\mathbb{E}_{P_N} W_N \rightarrow 1$, and the probability law with derivative $W_N/\mathbb{E}_{P_N} W_N$ converges to P_N in total variation. Conversely, total-variation convergence of laws with derivatives R_N implies $R_N \rightarrow 1$ in P_N -probability. In particular, if $0 \leq W_N \leq 1$ and $W_N \rightarrow 1$ in probability, the first assertion applies.*

Proof. Writing $z_N = \mathbb{E} W_N$, we have $|z_N - 1| \leq \mathbb{E} |W_N - 1|$ and

$$\frac{1}{2} \mathbb{E} \left| \frac{W_N}{z_N} - 1 \right| \leq \frac{\mathbb{E} |W_N - 1| + |z_N - 1|}{2z_N}.$$

The converse follows from Markov's inequality and (2.5). The last assertion follows from boundedness and convergence in probability. $\square$

We write $u_N \gg v_N$ when $u_N/v_N \rightarrow \infty$, and $u_N \asymp v_N$ when their ratio is bounded above and below by positive constants. Asymptotic scales refer to fixed energy units; in dimensional form, ratios such as b_N^2/c_N or $\Delta_N s_N/c_N$ carry a factor β_N^2 , which changes no asymptotic condition.

2.3 Centered and secant calibrations

The centered calibration places the maximum of the reservoir likelihood L_N at one relevant energy center a_N :

$$\mathcal{E}_N = a_N + \frac{c_N}{\beta_N}. \quad (2.7)$$

After division by $L_N(a_N)$, the weight becomes

$$W_N(e) = \begin{cases} \exp[\beta_N(e - a_N)] \left(1 - \frac{\beta_N(e - a_N)}{c_N}\right)^{c_N}, & \beta_N(e - a_N) < c_N, \\ 0, & \beta_N(e - a_N) \geq c_N. \end{cases} \quad (2.8)$$

Since $\log(1 - y) \leq -y$ for every $y < 1$,

$$0 \leq W_N(e) \leq 1 \quad (e \in \mathbb{R}). \quad (2.9)$$

Because this bound covers arbitrarily low energies, the sufficiency part of Theorem 3.1 needs no moment or tail assumptions.

The secant calibration instead equalizes L_N at two centers $e_{-,N} < e_{+,N}$ with gap $\Delta_N = e_{+,N} - e_{-,N}$:

$$\mathcal{E}_N = e_{-,N} + \frac{\Delta_N}{1 - e^{-\beta_N \Delta_N / c_N}} = e_{+,N} + \frac{\Delta_N}{e^{\beta_N \Delta_N / c_N} - 1}. \quad (2.10)$$

Then $L_N(e_{-,N}) = L_N(e_{+,N}) > 0$. On the feasible half-line, let $h_N = \log[L_N/L_N(e_{-,N})]$ be the residual log-weight,

$$h_N(e) = \beta_N(e - e_{-,N}) + c_N \log \frac{\mathcal{E}_N - e}{\mathcal{E}_N - e_{-,N}}, \quad e < \mathcal{E}_N, \quad (2.11)$$

and set $e^{h_N(e)} = 0$ at and above the cutoff. Thus $dQ_N/dP_N = e^{h_N(E_N)}/\mathbb{E}_{P_N} e^{h_N(E_N)}$ for the secant calibration, and

$$h_N(e_{-,N}) = h_N(e_{+,N}) = 0, \quad h'_N(e) = \beta_N - \frac{c_N}{\mathcal{E}_N - e}, \quad h''_N(e) = -\frac{c_N}{(\mathcal{E}_N - e)^2}. \quad (2.12)$$

Hence $h_N \geq 0$ between the centers and $h_N \leq 0$ outside them whenever feasible. Balancing the center values need not balance the integrated probabilities of two broad phases.

Lemma 2.3 (Bounds for the secant calibration). *Suppose $\Delta_N \rightarrow \infty$ and $c_N/\Delta_N \rightarrow \infty$. For all large N , with a constant C depending only on fixed bounds for β_N ,*

$$|h'_N(e_{\pm,N})| \leq C\Delta_N/c_N, \quad (2.13)$$

$$\sup_{e < \mathcal{E}_N} h_N(e) \leq C\Delta_N^2/c_N, \quad (2.14)$$

$$h_N(e) \leq C\frac{\Delta_N}{c_N}|e - e_{i,N}| \quad (i \in \{-, +\}, e < \mathcal{E}_N). \quad (2.15)$$

If $s_N \rightarrow \infty$, $s_N = o(\Delta_N)$, and $c_N \gg \Delta_N s_N$, then for each fixed R ,

$$\sup_{|e - e_{i,N}| \leq R s_N} |h_N(e)| \rightarrow 0, \quad (2.16)$$

and these windows lie below the cutoff for all sufficiently large N .

Proof. Put $t_N = \beta_N \Delta_N / c_N \rightarrow 0$. The residual reservoir energies at the lower and upper centers, $\Delta_N/(1 - e^{-t_N})$ and $\Delta_N/(e^{t_N} - 1)$, give inverse temperatures

$$\beta_N \frac{1 - e^{-t_N}}{t_N} \quad \text{and} \quad \beta_N \frac{e^{t_N} - 1}{t_N}.$$

Their difference from β_N is $O(t_N)$, giving (2.13). Between the centers, the residual reservoir energy is $(c_N/\beta_N)[1 + O(t_N)]$, so $0 \leq -h''_N \leq C/c_N$. Interpolation between the two zero endpoints gives

$$0 \leq h_N(e) \leq \frac{C}{2c_N}(e - e_{-,N})(e_{+,N} - e).$$

(Subtracting the right side from h_N leaves a convex function vanishing at both endpoints, hence nonpositive on the interval.) Outside the interval, concavity gives $h_N \leq 0$. This proves (2.14). A concave differentiable function lies below each tangent, proving (2.15).

For (2.16), the cutoff lies at distance asymptotic to c_N/β_N from either center, so $-h''_N \leq C/c_N$ on every indicated window for large N . Taylor's theorem and the endpoint slope estimate give

$$|h_N(e)| \leq C \frac{\Delta_N R s_N + R^2 s_N^2}{c_N} \rightarrow 0.$$

□

3 Sharp reservoir scales

We treat three energy locations on one scale, then two narrow phases separated by a larger gap.

3.1 Three support points determine a quadratic threshold

Theorem 3.1 (Weak-support criterion). *Suppose that, under the canonical reference law,*

$$\frac{E_N - a_N}{b_N} \Rightarrow X, \quad b_N \rightarrow \infty, \quad (3.1)$$

and that the law of X has at least three distinct points in its support. For any prescribed exponents $c_N > 0$,

$$[\exists \text{ admissible } \mathcal{E}_N : d_{\text{TV}}(Q_N, P_N) \rightarrow 0] \iff c_N \gg b_N^2. \quad (3.2)$$

The centered calibration (2.7) suffices, and the necessity holds for every admissible calibration.

Necessity rests on the following chord inequality.

Lemma 3.2 (Three-energy chord defect). *For $0 < \theta < 1$ and $q > 0$, define*

$$f_\theta(q) = \log[(1 - \theta)e^q + \theta] - (1 - \theta)q. \quad (3.3)$$

Then

$$f_\theta(q) \geq \frac{\theta(1 - \theta)}{2e} \min\{q^2, q\}. \quad (3.4)$$

If $g(e) = \beta_N e + c_N \log(\mathcal{E}_N - e)$ and $e_1 < e_2 < e_3 < \mathcal{E}_N$, set

$$D = e_3 - e_1, \quad \theta = \frac{e_2 - e_1}{D}, \quad q = \log \frac{\mathcal{E}_N - e_1}{\mathcal{E}_N - e_3}. \quad (3.5)$$

The exact identities are

$$g(e_3) - g(e_1) = \beta_N D - c_N q, \quad (3.6)$$

$$g(e_2) - (1 - \theta)g(e_1) - \theta g(e_3) = c_N f_\theta(q). \quad (3.7)$$

Proof. We have $f_\theta(0) = f'_\theta(0) = 0$ and

$$f''_\theta(q) = \frac{\theta(1 - \theta)e^q}{[(1 - \theta)e^q + \theta]^2}.$$

For $0 \leq q \leq 1$, the denominator is at most e^{2q} , giving $f''_\theta(q) \geq e^{-1}\theta(1 - \theta)$. Integrating twice proves (3.4) for $q \leq 1$. Convexity and $f_\theta(0) = 0$ imply that $f_\theta(q)/q$ is nondecreasing for $q > 0$, which proves it for $q \geq 1$. For the identities, the linear energy term cancels in the chord and $\mathcal{E}_N - e_2 = (1 - \theta)(\mathcal{E}_N - e_1) + \theta(\mathcal{E}_N - e_3)$. Substitution gives both formulas. $\square$

Proof of Theorem 3.1. For sufficiency, use the centered weight (2.8). Weak convergence implies $E_N - a_N = O_{P_N}(b_N)$. If $c_N \gg b_N^2$, the event $|\beta_N(E_N - a_N)/c_N| \leq 1/2$ has probability tending to one, and on it Taylor's theorem gives

$$\log W_N(E_N) = -\frac{\beta_N^2(E_N - a_N)^2}{2c_N} \left[1 + O\left(\frac{|E_N - a_N|}{c_N}\right) \right] \rightarrow 0$$

in probability. The uniform bound (2.9) and Lemma 2.2 prove convergence and admissibility for all large N ; finitely many N do not matter.

For necessity, suppose some admissible calibration gives convergence; write $R_N = L_N/Z_N$. Choose $\varepsilon_N \downarrow 0$ slowly enough that $P_N(|R_N - 1| > \varepsilon_N) \rightarrow 0$. On the complementary good set, $R_N > 0$ and $|\log R_N| \leq 2\varepsilon_N$ for all large N , so every good energy lies strictly below the cutoff.

Choose three bounded open intervals $I_1 < I_2 < I_3$, with positive gaps, around three distinct support points of X . By the Portmanteau theorem each has positive limiting lower probability, so each event $\{(E_N - a_N)/b_N \in I_j\}$ intersects the good set for all large N ; select an energy $e_{j,N}$ in each intersection. The parameters in (3.5) obey $D_N \asymp b_N$ and $\theta_N \in [\delta, 1 - \delta]$ for some fixed $\delta > 0$. The three log likelihoods are $o(1)$, so (3.6)–(3.7) imply

$$c_N q_N = \beta_N D_N + o(1), \quad c_N f_{\theta_N}(q_N) = o(1). \quad (3.8)$$

Since $D_N \asymp b_N \rightarrow \infty$, the first quantity diverges. The chord lower bound then forces $q_N \rightarrow 0$: otherwise $c_N \min(q_N^2, q_N)$ is bounded below along a subsequence by a positive multiple of $c_N q_N \rightarrow \infty$. Since $q_N \leq 1$ for large N , the same bound gives $c_N q_N^2 \rightarrow 0$. Finally,

$$\frac{D_N^2}{c_N} = \frac{c_N q_N^2}{\beta_N^2} [1 + o(1)] \rightarrow 0.$$

Since $D_N \asymp b_N$, this is the claimed necessity. $\square$

Proposition 3.3 (The two-atom exception). *If the canonical energy law has exactly two atoms $e_- < e_+$, with any positive probabilities, then for every $c_N > 0$ the choice (2.10) reproduces the complete canonical state law exactly.*

Proof. The residual weight has equal positive values at both atoms, so its normalized value is one P_N -almost surely. Apply Lemma 2.1. $\square$

A two-atom *scaling limit* is not an exactly two-atom law: fluctuations hidden by the coarse scaling can impose a larger requirement than Proposition 3.3 reveals.

3.2 Two phases with unequal energy scales

When the gap Δ_N greatly exceeds the fluctuation scale s_N , fluctuations within one phase supply the third energy for the chord argument. Phase measures here and below are positive probability measures, possibly overlapping or defined through phase labels on an enlarged probability space, provided E_N remains the actual subsystem energy.

Theorem 3.4 (Two-scale necessity). *Suppose the energy law admits the positive decomposition*

$$P_N^E = w_{-,N} P_{-,N} + w_{+,N} P_{+,N} + \delta_N R_N^{\text{exc}}, \quad w_{-,N} + w_{+,N} + \delta_N = 1, \quad (3.9)$$

where $\liminf_N w_{i,N} > 0$ for $i = -, +$. Let $e_{-,N} < e_{+,N}$ be phase centers with $\Delta_N = e_{+,N} - e_{-,N} \rightarrow \infty$, and let $s_N \rightarrow \infty$ with $s_N = o(\Delta_N)$. In at least one phase, say the lower phase,

$$(E_N - e_{-,N})/s_N \Rightarrow X_- \quad \text{under } P_{-,N}, \quad (3.10)$$

where $\text{supp } X_-$ has at least two distinct points. In the other phase, assume $(E_N - e_{+,N})/\Delta_N \rightarrow 0$ in probability. Then any admissible calibration for which $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ must satisfy

$$c_N \gg \Delta_N s_N. \quad (3.11)$$

The same assertion holds with the phase roles interchanged.

Proof. The good set from the previous proof has probability tending to one under each phase law, since its complementary phase probability is at most the unconditional one divided by $w_{i,N}$. Two bounded separated neighborhoods of support points of X_- supply good energies $e_{1,N} < e_{2,N}$ with $e_{2,N} - e_{1,N} \asymp s_N$ and $e_{j,N} = e_{-,N} + O(s_N)$. Concentration in the upper phase supplies a good energy $e_{3,N} = e_{+,N} + o(\Delta_N)$; explicitly, choose a deterministic sequence $r_N \downarrow 0$ slowly enough that $P_{+,N}(|E_N - e_{+,N}| \leq r_N \Delta_N) \rightarrow 1$. Thus $D_N \asymp \Delta_N$ and $\theta_N(1 - \theta_N) \asymp s_N/\Delta_N$.

The identities (3.8) still hold. Since β_N is bounded above and away from zero and $D_N \rightarrow \infty$, they imply $q_N \asymp \Delta_N/c_N$. The chord bound is therefore at least a positive constant times

$$c_N \frac{s_N}{\Delta_N} \min(q_N^2, q_N) \asymp \min \left\{ \frac{\Delta_N s_N}{c_N}, s_N \right\}.$$

It must tend to zero; since $s_N \rightarrow \infty$, this forces $\Delta_N s_N/c_N \rightarrow 0$. If the upper phase provides the two points, use $1 - \theta_N \asymp s_N/\Delta_N$ instead. $\square$

The obstruction uses only positive-probability neighborhoods of the two phases, not the interphase valley.

3.3 A positive-phase sufficient criterion

The converse needs more information, because the secant calibration can amplify rare energies between the peaks by up to $\exp[O(\Delta_N^2/c_N)]$ (Lemma 2.3).

Theorem 3.5 (Positive decomposition and exceptional mass). *Assume the positive decomposition (3.9), with $\Delta_N \rightarrow \infty$, $s_N \rightarrow \infty$, and $s_N = o(\Delta_N)$. Suppose that, for some fixed $\eta > 0$, $M < \infty$, and N_0 ,*

$$\sup_{N \geq N_0, i \in \{-, +\}} \mathbb{E}_{P_{i,N}} \exp \left(\eta \frac{|E_N - e_{i,N}|}{s_N} \right) \leq M. \quad (3.12)$$

There is a fixed constant C depending only on bounds for β_N such that the conditions

$$c_N \gg \Delta_N s_N, \quad \delta_N \exp(C \Delta_N^2/c_N) \rightarrow 0 \quad (3.13)$$

ensure $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ with the secant calibration (2.10). The phase weights need not have positive lower limits.

Proof. Let $W_N = e^{h_N}$, zero at and above the cutoff; Lemma 2.3 applies since $c_N/\Delta_N \rightarrow \infty$. The exponential moment gives tightness on the phase scale, so (2.16) yields $W_N \rightarrow 1$ in $P_{i,N}$ -probability for each phase. The tangent bound gives globally

$$W_N(e) \leq \exp \left(\epsilon_N \frac{|e - e_{i,N}|}{s_N} \right), \quad \epsilon_N = C \Delta_N s_N/c_N \rightarrow 0.$$

For large N , $2\epsilon_N \leq \eta$, so $\mathbb{E}_{P_{i,N}} W_N^2 \leq M$; by uniform integrability, $\mathbb{E}_{P_{i,N}} |W_N - 1| \rightarrow 0$ for each phase. On the exceptional component,

$$\delta_N \mathbb{E}_{R_N^{\text{exc}}} |W_N - 1| \leq \delta_N \{1 + \exp(C \Delta_N^2/c_N)\} \rightarrow 0.$$

Adding the positive components and using Lemma 2.2 completes the proof. $\square$

Corollary 3.6 (A sharp two-phase criterion). *Assume the positive decomposition, scale hypotheses, and moment bound of Theorem 3.5. Suppose additionally that the phase weights have positive lower limits, at least one scaled phase law has a weak limit with two distinct support points, and*

$$\delta_N \leq \exp(-b\Delta_N/s_N) \quad \text{for some fixed } b > 0. \quad (3.14)$$

Then a calibration with $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ exists if and only if $c_N \gg \Delta_N s_N$. In particular, if $\Delta_N \asymp N$ and $s_N \asymp \sqrt{N}$, the sharp scale is $c_N \gg N^{3/2}$.

Proof. For necessity, the moment bound makes the other phase tight on scale s_N , hence concentrated on the smaller-than-gap scale required by Theorem 3.4. For sufficiency, $\Delta_N^2/c_N = o(\Delta_N/s_N)$ and $\Delta_N/s_N \rightarrow \infty$, so (3.14) implies the second condition in (3.13). $\square$

More generally, if $\delta_N \leq e^{-t_N}$ with $t_N \rightarrow \infty$, the sufficient condition

$$c_N \gg \max\{\Delta_N s_N, \Delta_N^2/t_N\} \quad (3.15)$$

follows. As a condition on the exceptional mass alone it is not necessary, since an exceptional component near the phase centers may be amplified much less than the global maximum used in the proof. The exceptional part must be the positive measure $\delta_N R_N^{\text{exc}}$, not a signed remainder in a partition-function expansion, and the exponential moments must concern the energy actually exchanged with the reservoir.

3.4 An alternative criterion allowing interfacial tails

Theorem 3.5 requires a positive decomposition with uniform phase exponential moments and an exponentially small exceptional mass. The criterion below replaces both by one pointwise bound on the energy density between the phases: Gaussian near each center and interfacial, with cost x^α at distance x , farther away, as for a droplet surface. For $\alpha > 1/2$ it also gives the boundary law of Section 6 for every capacity coefficient. Let p_N be the density of P_N^E with respect to Lebesgue measure and let φ_v denote the centered Gaussian density of variance v . Assume

$$\Delta_N/N \rightarrow \ell > 0, \quad (3.16)$$

$$\sqrt{N} p_N(e_{i,N} + \sqrt{N}z) \rightarrow w_i \varphi_{v_i}(z) \quad \text{in } L^1([-R, R]) \quad \text{for every } R < \infty, \quad (3.17)$$

where $w_i > 0$, $w_- + w_+ = 1$, and $v_i > 0$. Between the centers, set $x_N(e) = \min(e - e_{-,N}, e_{+,N} - e)$. Suppose fixed $A, a > 0$ and $\alpha \in [1/2, 1]$ give

$$p_N(e) \leq \frac{A}{\sqrt{N}} \exp \left[-a \min \left\{ \frac{x_N(e)^2}{N}, x_N(e)^\alpha \right\} \right], \quad e \in [e_{-,N}, e_{+,N}]. \quad (3.18)$$

The minimum interpolates between the Gaussian cost $x_N(e)^2/N$ and the interfacial cost $x_N(e)^\alpha$, which grows at most linearly; no envelope is imposed outside the phase interval.

We prove sufficiency for a general concave residual log-weight, which also covers the smooth reservoirs of Section 6. Let I_N be an interval of feasible energies that contains the two centers and all their fixed $R\sqrt{N}$ neighborhoods for large N , let h_N be concave and twice continuously differentiable in its interior, and set $e^{h_N} = 0$ outside I_N . Suppose

$$h_N(e_{-,N}) = h_N(e_{+,N}) = 0, \quad 0 \leq -h_N''(e) \leq K\kappa_N \quad (3.19)$$

between the centers and on each such neighborhood, with a constant K independent of N and R .

Proposition 3.7 (Sufficiency with an interior density envelope). *Under (3.16)–(3.19), if $\kappa_N N^{3/2} \rightarrow 0$, then*

$$\mathbb{E}_{P_N} |e^{h_N(E_N)} - 1| \rightarrow 0. \quad (3.20)$$

The normalized reweighted law therefore converges in total variation.

Proof. Integrating h'_N between the zero endpoints and using its $K\kappa_N$ Lipschitz bound gives $|h'_N(e_{i,N})| \leq K\kappa_N \Delta_N$. Hence, uniformly for $|z| \leq R$,

$$|h_N(e_{i,N} + \sqrt{N}z)| \leq K\kappa_N \Delta_N \sqrt{N}R + \frac{1}{2}K\kappa_N N R^2 \rightarrow 0. \quad (3.21)$$

Interpolation also gives, between the centers,

$$0 \leq h_N(e) \leq \frac{1}{2}K\kappa_N(e - e_{-,N})(e_{+,N} - e) \leq C\kappa_N N x_N(e). \quad (3.22)$$

Set $\lambda_N = \kappa_N N^{3/2}$. For $R\sqrt{N} \leq x \leq \Delta_N/2$, the ratios of the last bound to the two costs in (3.18) satisfy

$$\frac{C\kappa_N N x}{x^2/N} \leq C\lambda_N/R, \quad \frac{C\kappa_N N x}{x^\alpha} \leq C\lambda_N N^{1/2-\alpha}.$$

Both tend to zero for each fixed $R > 0$, so for large N the reservoir-weighted interior mass outside the phase windows is at most

$$\frac{2A}{\sqrt{N}} \int_{R\sqrt{N}}^{\Delta_N/2} \exp\left[-\frac{a}{2} \min\{x^2/N, x^\alpha\}\right] dx. \quad (3.23)$$

Since $e^{-a \min(u,v)/2} \leq e^{-au/2} + e^{-av/2}$, the Gaussian part has limiting upper bound $2A \int_R^\infty e^{-az^2/2} dz$. The other part tends to zero for fixed $R > 0$: its integral over the positive half-line is finite, and it carries an extra factor $N^{-1/2}$.

Outside the phase interval, concavity gives $h_N \leq 0$ where feasible and the weight vanishes elsewhere, so these tails are not amplified. Equation (3.17), with the weights summing to one, implies

$$\lim_{R \rightarrow \infty} \limsup_{N \rightarrow \infty} P_N \left(E_N \notin \bigcup_i [e_{i,N} - R\sqrt{N}, e_{i,N} + R\sqrt{N}] \right) = 0.$$

On each fixed phase window, (3.21) makes the weight uniformly approach one. On the remaining interior, use (3.23) and the corresponding unweighted tail; on the exterior, use $|e^{h_N} - 1| \leq 1$. Taking $N \rightarrow \infty$ first and then $R \rightarrow \infty$ proves (3.20). $\square$

Corollary 3.8 (Two-phase threshold with interfacial tails). *Under (3.16)–(3.18), an exact reservoir (2.2) can reproduce the canonical law in total variation by some calibration if and only if $c_N \gg N^{3/2}$.*

Proof. For sufficiency, choose (2.10). If $c_N \gg N^{3/2}$, the residual reservoir energy is $(c_N/\beta_N)[1+o(1)]$ uniformly between the centers and in each fixed phase neighborhood. Thus (3.19) holds with $\kappa_N = 1/c_N$, and Proposition 3.7 applies.

For necessity, divide the energy line at the midpoint of the centers. The local limits and their total weight one imply that the two resulting positive conditional phase measures have weights tending to w_i and scaled energy laws converging weakly to $\mathcal{N}(0, v_i)$: their mass in every fixed scaled compact interval has the stated limit, and the limiting Gaussian mass tends to the entire phase weight as the interval grows, so no mass escapes on either side of the midpoint. Theorem 3.4 with $s_N = \sqrt{N}$ then gives the condition for any calibration. $\square$

The density hypotheses are stronger than local coexistence limits; they must be verified for the specified microscopic Hamiltonian, not inferred from two peaks. Proposition 3.7 concerns laws with a Lebesgue density, whereas Theorems 3.1, 3.4, and 3.5 apply without change to discrete, continuous, or mixed laws and need no local limit theorem.

4 Microscopic realizations

This section verifies the hypotheses of Section 3 for the mean-field Potts model, with exact finite-system formulas, and for the short-range Potts model, via contour estimates. In both, the reservoir threshold follows from the established phase structure; we make no new claim about the location or order of the transitions.

4.1 The three-state Curie–Weiss–Potts model

Let $n_1 + n_2 + n_3 = N$ be the occupation numbers of N three-state spins and take

$$U_N = -\frac{J}{2N} \sum_{j=1}^3 n_j^2, \quad J > 0, \quad \beta J = 4 \log 2. \quad (4.1)$$

Optionally add $2aN$ independent quadratic momentum coordinates, with $a > 0$ fixed and $2aN$ integral along the size sequence. Their canonical energy K_N is Gamma distributed with shape aN and rate β , and $E_N = U_N + K_N$. For $a = 0$ there are no momenta and $K_N = 0$.

Theorem 4.1 (Mean-field microscopic reservoir threshold). *For every fixed $J > 0$ and $a \geq 0$ in the model just defined, an admissible calibration for which $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ exists if and only if $c_N \gg N^{3/2}$. The secant calibration (2.10) uses the centers*

$$e_{-,N} = N \left(-\frac{J}{4} + \frac{a}{\beta} \right), \quad e_{+,N} = N \left(-\frac{J}{6} + \frac{a}{\beta} \right). \quad (4.2)$$

The mean-field phase structure and fluctuations are known in greater generality [12, 19]; we derive what is needed, making the normalization and the degenerate disordered spin-energy fluctuation explicit.

Write $p_j = n_j/N$, with coordinates (p_1, p_2) on the probability simplex and $p_3 = 1 - p_1 - p_2$. The rate potential, up to an additive constant, is

$$I(p) = \sum_{j=1}^3 p_j \log p_j - \frac{\beta J}{2} \sum_{j=1}^3 p_j^2. \quad (4.3)$$

Its minima are one disordered and three ordered points,

$$p^{(d)} = (1/3, 1/3, 1/3), \quad p^{(o)} \in \{(2/3, 1/6, 1/6), (1/6, 2/3, 1/6), (1/6, 1/6, 2/3)\}. \quad (4.4)$$

The list is complete. First, a boundary point cannot minimize I : transferring a small mass $\varepsilon > 0$ to a zero coordinate lowers the entropy term by order $\varepsilon |\log \varepsilon|$ and the quadratic term by only $O(\varepsilon)$. At an interior stationary point, all $\log p_j - \beta J p_j$ are equal. Since $\log x - \beta J x$ is strictly concave, the coordinates take at most two values. If two coordinates share the larger value x and the remaining value is $y < x$, stationarity gives $\beta J = \log(x/y)/(x - y) > 1/x$, and the second variation exchanging

the two large coordinates is negative. A nonsymmetric minimum must therefore have one large coordinate $r > 1/3$ and two equal small coordinates $(1-r)/2$. Its stationary equation is

$$\log \frac{2r}{1-r} - \frac{\beta J}{2}(3r-1) = 0. \quad (4.5)$$

The derivative of the left side is $1/[r(1-r)] - 3\beta J/2$, which has at most two zeros. By Rolle's theorem, (4.5) has at most three roots; they are $1/3, 1/2, 2/3$. At $r = 1/2$ the second variation along this curve is $4 - 6 \log 2 < 0$. The other candidates have the same value $I_{\min} = -\log 3 - (2/3) \log 2$, and their Hessians are positive definite:

$$H_d = (3 - \beta J) \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \det H_d = 3(3 - \beta J)^2, \quad (4.6)$$

$$H_o = \begin{pmatrix} 15/2 - 2\beta J & 6 - \beta J \\ 6 - \beta J & 12 - 2\beta J \end{pmatrix}, \quad \det H_o = 3(3 - \beta J)(6 - \beta J). \quad (4.7)$$

Partition the simplex into the four Voronoi cells of these minima (ties resolved deterministically). All four cells have positive limiting probabilities. Uniform Stirling expansion on compact interior sets gives the unnormalized occupation weight

$$\frac{N!}{\prod_j n_j!} e^{-\beta U_N(n)} = \frac{1 + o(1)}{2\pi N \sqrt{p_1 p_2 p_3}} e^{-NI(p)}. \quad (4.8)$$

Near a minimum, the rescaled two-dimensional lattice has mesh $N^{-1/2}$ and hence density N . Taylor expansion of I in (4.8) and Gaussian Riemann sums give a phase contribution proportional to $(p_1 p_2 p_3)^{-1/2} (\det H_i)^{-1/2}$ and a conditional occupation limit $\mathcal{N}(0, H_i^{-1})$; the uniform bound proved below controls the tails. With minus denoting the combined ordered phases, the phase weights are

$$r_* := \sqrt{\frac{2(3 - \beta J)}{6 - \beta J}}, \quad w_- := \frac{3r_*}{1 + 3r_*}, \quad w_+ := \frac{1}{1 + 3r_*}. \quad (4.9)$$

The tangent gradient of $u(p) = -J \sum_j p_j^2/2$ is $(-J/2, 0)$ at the ordered minimum $p^{(o)} = (2/3, 1/6, 1/6)$ and zero at the disordered one. The delta method and independence of the kinetic energy give the conditional weak energy limits

$$\frac{E_N - e_{i,N}}{\sqrt{N}} \Rightarrow \mathcal{N}(0, v_i), \quad v_- := \frac{a}{\beta^2} + \frac{J^2}{6(3 - \beta J)}, \quad v_+ := \frac{a}{\beta^2}. \quad (4.10)$$

For $a = 0$, $\mathcal{N}(0, 0)$ is the point mass at zero: the disordered potential energy fluctuates only on scale one, since its first derivative on the simplex vanishes.

Lemma 4.2 (Uniform positive-phase energy moments). *For each fixed $a \geq 0$, there are $\eta > 0$ and $M < \infty$ such that the two positive energy phase laws defined by the Voronoi partition satisfy*

$$\sup_{N \geq N_0, i \in \{-, +\}} \mathbb{E} \left[\exp \left(\eta \frac{|E_N - e_{i,N}|}{\sqrt{N}} \right) \middle| i \right] \leq M. \quad (4.11)$$

Here N_0 is fixed and large enough that every conditioning phase has positive probability.

Proof. We first prove a Gaussian bound for the occupation law. Let $\mathcal{M}$ be the set of the four minima. Compactness, the complete classification of minima, and their positive definite Hessians give $I(p) - I_{\min} \geq c_0 \text{dist}(p, \mathcal{M})^2$ globally. The Gaussian neighborhoods in (4.8) give a partition function lower bound $ce^{-NI_{\min}}$. On a fixed interior subset, Stirling bounds then give

$$P_N(n) \leq \frac{C}{N} e^{-cN \text{dist}(n/N, \mathcal{M})^2}.$$

It extends globally with a smaller c : on a boundary strip disjoint from $\mathcal{M}$, where the rate gap is at least $\zeta > 0$, the multinomial type bound $N! / \prod n_j! \leq e^{-N \sum p_j \log p_j}$ gives $P_N(n) \leq Ce^{-N[I(p) - I_{\min}]}$ there. Splitting the exponent between the gap and distance bounds recovers the missing $1/N$, since $Ne^{-N\zeta/2}$ is bounded.

In the cell of $p^{(i)}$, the distance to $\mathcal{M}$ is $|p - p^{(i)}|$ and the Lipschitz property of u gives $|U_N - Nu(p^{(i)})|/\sqrt{N} \leq C\sqrt{N}|p - p^{(i)}|$; completing the square in the lattice Gaussian sum shows

$$\frac{C}{N} \sum_{n_1, n_2} e^{-cN|n/N - p^{(i)}|^2 + \eta C\sqrt{N}|n/N - p^{(i)}|} \leq C_\eta.$$

Conditioning preserves it, as the phase probabilities are bounded below for large N . For $a > 0$, the centered Gamma transform is

$$\mathbb{E} e^{t(K_N - aN/\beta)/\sqrt{N}} = e^{-ta\sqrt{N}/\beta} \left(1 - \frac{t}{\beta\sqrt{N}}\right)^{-aN}.$$

It is uniformly bounded for fixed t and large N . Using $e^{t|x|} \leq e^{tx} + e^{-tx}$ and independence completes the proof. $\square$

Proof of Theorem 4.1. The gap of the two energy centers is $NJ/12$. Lemma 4.2 supplies a positive decomposition with no exceptional component and $\sqrt{N}$ moment bounds, and (4.10) a nondegenerate ordered weak limit ($v_- > 0$ because $\beta J < 3$). Apply Corollary 3.6. $\square$

4.2 Continuous energy limits and exact finite-system formulas

For $a > 0$ the mean-field law satisfies the density hypotheses of Section 3.4; this subsection proves this and derives the exact finite- N formulas used in Section 7.

For $a > 0$, the standardized Gamma density converges uniformly to the Gaussian density of variance a/β^2 , by Stirling's formula on compact standardized intervals and, outside them, unimodality: once the mode is inside, the supremum outside is at the nearest endpoint, and enlarging the interval bounds both tails uniformly. Convolution with the conditional spin-energy weak limits (the $a = 0$ case of (4.10)) therefore gives the local L^1 density limits (3.17) with weights (4.9), and the stronger interior bound

$$p_N(E) \leq \frac{C}{\sqrt{N}} \sum_{i \in \{-, +\}} \exp \left[-c \frac{(E - e_{i,N})^2}{N} \right] \quad (e_{-,N} \leq E \leq e_{+,N}). \quad (4.12)$$

To prove it, note that the kinetic energies allowed in this interval obey $0 < K \leq C_0 N$, where the kinetic density g_N has logarithmic curvature $-(aN - 1)/K^2 \leq -c_1/N$ for all large N , and its maximum is $O(N^{-1/2})$. Expanding at its mode, which differs from aN/β by a constant, gives $g_N(K) \leq CN^{-1/2} e^{-c(K - aN/\beta)^2/N}$ on this interval. For a nearest minimum $p^{(i)}$, Lipschitz continuity of u yields

$$|p - p^{(i)}|^2 + |E/N - u(p) - a/\beta|^2 \geq c\{|p - p^{(i)}|^2 + |E/N - u(p^{(i)}) - a/\beta|^2\}.$$

Multiply the occupation bound from Lemma 4.2 by the Gamma bound and sum over each cell; the lattice Gaussian sum is $O(N)$ and cancels the prefactor N^{-1} , proving (4.12). This also controls contamination from the other phase in the local density limit.

For finite systems set $A = aN > 0$ and $m(n) = N!/(n_1!n_2!n_3!)$. The canonical and reservoir occupation laws are exactly

$$P_N(n) = \frac{m(n)e^{-\beta U_N(n)}}{\sum_{n'} m(n')e^{-\beta U_N(n')}}, \quad (4.13)$$

$$Q_N(n) = \frac{m(n)[\mathcal{E}_N - U_N(n)]_+^{A+c_N}}{\sum_{n'} m(n')[\mathcal{E}_N - U_N(n')]_+^{A+c_N}}. \quad (4.14)$$

Conditional on n , K_N is canonical Gamma(A, β), whereas

$$\frac{K_N}{\mathcal{E}_N - U_N(n)} \Big| n \sim \text{Beta}(A, c_N + 1) \quad \text{under } Q_N. \quad (4.15)$$

Equations (4.14) and (4.15) follow from integrating the product of subsystem and reservoir densities:

$$\int_0^{\mathcal{E}_N - U_N(n)} K^{A-1} [\mathcal{E}_N - U_N(n) - K]^{c_N} dK = B(A, c_N + 1) [\mathcal{E}_N - U_N(n)]^{A+c_N}.$$

For $a = 0$ the occupation formulas hold with $A = 0$ and $K_N = 0$; no shape-zero Gamma or Beta law is used.

Let G_A be the CDF of a unit-rate Gamma law and $I_x(A, c_N + 1)$ the regularized incomplete Beta function, extended by zero for $x \leq 0$ and one for $x \geq 1$. Then, for $a > 0$,

$$P_N(E_N \leq t) = \sum_n P_N(n) G_A(\beta[t - U_N(n)]), \quad (4.16)$$

$$Q_N(E_N \leq t) = \sum_{n: Q_N(n) > 0} Q_N(n) I_{[t - U_N(n)]/[\mathcal{E}_N - U_N(n)]}(A, c_N + 1), \quad (4.17)$$

where $G_A(x) = 0$ for $x \leq 0$. These are exact finite sums. The log likelihood in (2.3) is strictly concave below its cutoff and tends to $-\infty$ at both ends of that half-line; for $a > 0$ it is not constant on the canonical support, so its normalized maximum exceeds zero. The normalized log likelihood is therefore positive exactly on an interval (t_L, t_R) between its two roots, and

$$d_{\text{TV}}(Q_N, P_N) = Q_N(t_L < E_N < t_R) - P_N(t_L < E_N < t_R). \quad (4.18)$$

Equations (4.16)–(4.17) evaluate this directly, even when an endpoint lies outside the subsystem support. For the pure-spin model one instead sums the positive likelihood differences over the discrete occupation law.

4.3 A short-range Potts theorem

Fix a dimension $d \geq 2$ and an integer $q \geq q_0(d)$, independent of system size; the threshold $q_0(d) > 2d$ is fixed at the start of Appendix A, large enough for the cited contour results. On the periodic torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^d$, with $N = L^d$ and the set $\mathcal{B}_L$ of its dN nearest-neighbor bonds, let

$$U_N(\sigma) = - \sum_{\{x,y\} \in \mathcal{B}_L} \mathbf{1}_{\{\sigma_x = \sigma_y\}}, \quad \sigma \in \{1, \dots, q\}^{\Lambda_L}. \quad (4.19)$$

The unit coupling fixes energy units, and β_c denotes the coexistence inverse temperature.

Theorem 4.3 (Short-range microscopic reservoir threshold). *For every fixed $d \geq 2$ and fixed $q \geq q_0(d)$, the canonical law of (4.19) at β_c can be reproduced in total variation by an exact reservoir (2.2), with some calibration, if and only if $c_N \gg N^{3/2}$. The sufficient choice is (2.10) with centers Nu_o and Nu_d , where $u_o < u_d$ are the two coexistence energy densities.*

In contrast to the boundary results of Section 6, this threshold holds without restriction in $d = 2$, for the plain spin model. Its proof in Appendix A uses two positive phase partitions that need not coincide: midpoint energy events for necessity, through Theorem 3.4 with gap $N(u_d - u_o)$ and $s_N = \sqrt{N}$, and contour events in the Edwards–Sokal joint space of spins and bonds for sufficiency, through Theorem 3.5 and projection to the spins. Contour estimates control $\sqrt{N}$ -scale exponential moments of the occupied-bond count, but sufficiency needs them for the spin energy. The conditional binomial law of bonds given spins provides this transfer; differentiating a bond-conditioned spin law would not justify it. The exceptional contour mass decays on the surface scale L^{d-1} , which dominates the largest reservoir gain when $c_N \gg N^{3/2}$.

5 Canonical marginals and correlated phases under a shared reservoir

This section proves that, for $N \ll c_N \ll N^2$, a reservoir shared by two identical copies under the centered calibration selects opposite phases (Theorem 5.1) and that the mutual information of the two full states tends to one bit (Proposition 5.2). With balanced phase weights, each one-copy marginal is asymptotically canonical while the joint TV tends to $1/2$. Joint canonical accuracy requires $c_N \gg N^2$ (Corollary 5.3). Section 5.5 balances the phase weights with a temperature shift of order $1/N$. Thermally coupled systems with negative specific heat were shown to develop different subsystem phases and to exchange their assignments [29]; we give the reservoir-size conditions for this behavior in the full microscopic law.

5.1 Opposite-phase selection by the exact reservoir

Let P_N be the canonical law of one copy of size N at the specified inverse temperature $\beta_N \rightarrow \beta > 0$. Assume two disjoint measurable phase events $A_{-,N}, A_{+,N}$ partition its state space, with

$$w_{i,N} := P_N(A_{i,N}) \longrightarrow w_i > 0, \quad w_- + w_+ = 1. \quad (5.1)$$

Let $e_{-,N} < e_{+,N}$ be deterministic energy centers, with $\Delta_N = e_{+,N} - e_{-,N}$, chosen so that

$$\Delta_N/N \longrightarrow \ell > 0, \quad (E_N - e_{i,N})/\sqrt{N} \mid A_{i,N} \text{ is tight.} \quad (5.2)$$

These assumptions allow discrete phase laws and a degenerate conditional energy limit.

Two noninteracting copies, with full microscopic states X_1, X_2 and phase labels S_1, S_2 , have the reference law $\Pi_N = P_N \otimes P_N$. Their combined energy $E_N(X_1) + E_N(X_2)$ is coupled to one exact reservoir with the centered calibration

$$M_N = e_{-,N} + e_{+,N}, \quad \mathcal{E}_N = M_N + c_N/\beta_N. \quad (5.3)$$

The centered weight from (2.8) is

$$W_N = \begin{cases} \exp[\beta_N T_N](1 - \beta_N T_N/c_N)^{c_N}, & \beta_N T_N < c_N, \\ 0, & \beta_N T_N \geq c_N, \end{cases} \quad T_N = E_N(X_1) + E_N(X_2) - M_N. \quad (5.4)$$

Let $z_N = \mathbb{E}_{\Pi_N} W_N$ and $dQ_N/d\Pi_N = W_N/z_N$. By (2.9), $0 \leq W_N \leq 1$ for all states.

Theorem 5.1 (Opposite phases and full marginal accuracy). *Under (5.1)–(5.2), suppose*

$$N \ll c_N \ll N^2. \quad (5.5)$$

For $O_N = \{S_1 \neq S_2\}$, the shared reservoir with the centered calibration (5.3) satisfies

$$d_{\text{TV}}(Q_N, \Pi_N(\cdot \mid O_N)) \longrightarrow 0. \quad (5.6)$$

Consequently, with $Q_N^{(j)}$ denoting either one-copy marginal,

$$d_{\text{TV}}(Q_N, \Pi_N) \longrightarrow w_-^2 + w_+^2, \quad (5.7)$$

$$d_{\text{TV}}(Q_N^{(j)}, P_N) \longrightarrow |w_- - 1/2|, \quad j = 1, 2. \quad (5.8)$$

Proof. On either opposite-phase assignment $T_N = O_{\Pi_N}(\sqrt{N})$ conditionally. The Taylor expansion $x + \log(1 - x) = -x^2/2 + O(|x|^3)$, together with $N/c_N \rightarrow 0$, gives $\log W_N \rightarrow 0$ in its conditional probability; the cutoff lies beyond every fixed fluctuation window. On the same-phase assignments, $T_N = \pm \Delta_N + O_{\Pi_N}(\sqrt{N})$. Because $c_N/N \rightarrow \infty$, the Taylor expansion still applies in conditional probability, but now

$$\log W_N = -\frac{\beta_N^2 \Delta_N^2}{2c_N} [1 + o_{\Pi_N}(1)] \longrightarrow -\infty.$$

Thus $W_N - \mathbf{1}_{O_N} \rightarrow 0$ in probability and, by boundedness, in $L^1(\Pi_N)$. In particular, $z_N \rightarrow p_O := 2w_-w_+ > 0$. Normalizing the two nonnegative weights proves (5.6).

Conditioning a probability measure on an event of probability p changes it by $1 - p$ in total variation, which gives (5.7). The exact marginal of $\Pi_N(\cdot \mid O_N)$ is

$$\frac{1}{2}P_N(\cdot \mid A_{-,N}) + \frac{1}{2}P_N(\cdot \mid A_{+,N}).$$

The two phase supports are disjoint, so its distance from P_N is $|w_{-,N} - 1/2|$. Total variation contracts under projection, proving (5.8). $\square$

With the copies in opposite phases, the reservoir absorbs only within-phase energy fluctuations. No density approximation, contour decomposition, or bound on the probability of a particular interfacial configuration is needed: tightness controls the atypical canonical mass, and the weight (5.4), bounded by one, cannot amplify it.

5.2 The full-state mutual information

Total variation does not control relative entropy on growing or continuous state spaces, so mutual information needs a separate argument. For $Q \ll P$, define $D_{\text{KL}}(Q \parallel P) = \int \log(dQ/dP) dQ$, with $0 \log 0 = 0$. Logarithms are natural; $\log 2$ nats is one bit. The mutual information of a joint law Q is $I_Q(X_1; X_2) = D_{\text{KL}}(Q \parallel Q^{(1)} \otimes Q^{(2)})$.

Proposition 5.2 (One bit in the full state law). *Under the hypotheses of Theorem 5.1,*

$$I_{Q_N}(X_1; X_2) \longrightarrow \log 2. \quad (5.9)$$

This limit is independent of the original positive phase weights.

Proof. Since $x \log x$ is continuous and bounded on $[0, 1]$, $\mathbb{E}_{\Pi_N} W_N \log W_N \rightarrow 0$ by the proof of Theorem 5.1. Hence

$$D_{\text{KL}}(Q_N \| \Pi_N) = z_N^{-1} \mathbb{E}_{\Pi_N} W_N \log W_N - \log z_N \rightarrow -\log(2w_- w_+). \quad (5.10)$$

Each marginal likelihood $r_{j,N} = dQ_N^{(j)} / dP_N$ is at most $1/z_N$, hence uniformly bounded for large N . By (5.6) it converges in $L^1(P_N)$ to the finite- N conditional marginal ratio

$$r_N^O = \frac{\mathbf{1}_{A_{-,N}}}{2w_{-,N}} + \frac{\mathbf{1}_{A_{+,N}}}{2w_{+,N}}.$$

Both ratios lie in a common compact interval of $[0, \infty)$, on which $r \log r$ is uniformly continuous, so convergence in probability implies convergence of its integrals:

$$D_{\text{KL}}(Q_N^{(j)} \| P_N) \rightarrow \frac{1}{2} \log \frac{1}{2w_-} + \frac{1}{2} \log \frac{1}{2w_+}. \quad (5.11)$$

All divergences are finite because the likelihood ratios are bounded, and the relative-entropy chain rule gives

$$I_{Q_N}(X_1; X_2) = D_{\text{KL}}(Q_N \| \Pi_N) - \sum_{j=1}^2 D_{\text{KL}}(Q_N^{(j)} \| P_N). \quad (5.12)$$

Substituting (5.10) and (5.11) proves (5.9). $\square$

This $\log 2$ comes entirely from the uniform, perfectly anticorrelated phase labels; in the window (5.5), within-phase fluctuations add nothing in the limit. At capacity proportional to N , under Gaussian phase fluctuations with positive variances, they do (Proposition D.4 in Appendix D).

5.3 Joint canonical accuracy requires a quadratic-size reservoir

For a fixed number of copies, some calibration achieves joint canonical accuracy exactly when $c_N \gg N^2$.

Corollary 5.3 (Optimized joint threshold). *Suppose only that, for some centers a_N , $(E_N - a_N)/N$ converges weakly to a law with exactly two distinct positive-weight atoms. For any fixed number of independent identical copies $m \geq 2$, a shared reservoir (2.2) can reproduce their full-state canonical product law in total variation, by some calibration, if and only if $c_N \gg N^2$.*

Proof. After a common centering, the total energy divided by N converges to the m -fold convolution of the two-atom law. It has $m + 1 \geq 3$ distinct support points, all of positive weight. Apply Theorem 3.1 with $b_N = N$ to the combined subsystem. Since m is fixed, using N rather than mN does not change the condition. $\square$

Under the assumptions of Section 5, $(E_N - e_{-,N})/N$ has limiting atoms $0, \ell$ with weights w_-, w_+ , so the corollary applies; for $m = 2$ the combined energy $[E_N(X_1) + E_N(X_2) - M_N]/N$ has atoms $-\ell, 0, \ell$ with weights $w_-^2, 2w_- w_+, w_+^2$. No central limit theorem within the phases is needed.

For the microscopic models, an isolated copy has the sharp threshold $c_N \gg N^{3/2}$, yet with balanced phase weights a shared reservoir with $c_N = N^{5/4}$ (too small for an isolated copy) gives canonical one-copy marginals, because the copies trade the latent heat with each other: when one is in the lower phase the other is in the upper one, so their total energy stays within $O(\sqrt{N})$ of M_N and the reservoir absorbs only within-phase fluctuations. Appendix D also treats a fixed phase count among m copies (Proposition D.1).

5.4 Phase events for the microscopic models

Both microscopic models satisfy (5.1)–(5.2). For the mean-field model of Section 4.1, take the ordered and disordered Voronoi cells; Lemma 4.2 gives conditional exponential moments on the $\sqrt{N}$ scale, hence tightness, with or without momenta. For the short-range model of Section 4.3, take the midpoint energy events $B_{o,N}$ and $B_{d,N}$ defined in Appendix A; their conditional standardized energies converge weakly by (A.13), hence are tight.

5.5 Balancing the reference phase weights

The microscopic models have unequal limiting phase weights at coexistence, but β_N may be any specified sequence, so the canonical reference can be taken at a nearby temperature. A shift of order $1/N$ changes the limiting phase weights but leaves the phase-conditional laws asymptotically unchanged in total variation.

Lemma 5.4 (A bounded-energy temperature shift). *Let the spin energy U_N/N be uniformly bounded and suppose the phase conditional laws at β_c concentrate at $u_- < u_+$, with positive limiting weights w_-, w_+ . For $\beta_N = \beta_c + \delta/N + o(N^{-1})$, the new limiting weight ratio is*

$$\frac{w_-^{(\delta)}}{w_+^{(\delta)}} = \frac{w_-}{w_+} \exp[\delta(u_+ - u_-)]. \quad (5.13)$$

The new spin law conditional on either phase approaches the old phase-conditional law in total variation. Hence any conditional weak fluctuation limit, with the same center and scale, persists.

Proof. The unnormalized likelihood $\exp[-(\beta_N - \beta_c)U_N]$ of the change of canonical measure is bounded above and below by positive constants uniformly in N , and on phase i it converges in probability to $e^{-\delta u_i}$. Bounded convergence gives the limiting phase normalizations and (5.13). After normalizing within either phase, the likelihood converges to one in L^1 , proving the conditional total-variation assertion. $\square$

Writing $\ell = u_+ - u_-$, asymptotic balance is obtained by

$$\beta_N = \beta_c - \frac{\log(w_-/w_+)}{N\ell} + o(N^{-1}). \quad (5.14)$$

The negative sign means warming when the ordered, lower-energy phase has excess weight. For the short-range model $w_-/w_+ = q$, giving the shift $-\log q/(N\ell)$; for the three-state mean-field model the ratio is $3r_*$ from (4.9) and $\ell = J/12$. Both spin Hamiltonians are bounded per particle, so Lemma 5.4 applies.

The shift also preserves the isolated-copy $N^{3/2}$ thresholds. For the short-range positive contour decomposition, retain the conditional bond kernel $K_{\beta_c}(\mathrm{d}A \mid \sigma)$ of the Edwards–Sokal law at β_c (it need not be the kernel at β_N) and define the auxiliary law

$$\hat{P}_N(\mathrm{d}\sigma, \mathrm{d}A) = P_{\beta_N}(\mathrm{d}\sigma)K_{\beta_c}(\mathrm{d}A \mid \sigma).$$

Its likelihood relative to the old joint law is $\mathrm{d}P_{\beta_N}/\mathrm{d}P_{\beta_c}$, bounded above and below by positive constants, so the fixed contour events still give a positive decomposition of the new physical spin law; their conditional exponential moments remain bounded, and their exceptional mass increases by at most a constant factor. For the mean-field model, the same bounded spin likelihood directly preserves the positive phase moment bounds.

Independent momenta, taken at the same β_N , contribute a partition factor common to both spin phases, which cancels from the weight ratio. Their law has relative entropy with respect to the kinetic law at β_c (from the product Gaussian momenta or their Gamma energy law)

$$aN \left[\frac{\beta_c}{\beta_N} - 1 - \log \frac{\beta_c}{\beta_N} \right] = O(N^{-1}).$$

By Pinsker's inequality the two kinetic laws converge in total variation; the mean energy changes by $O(1)$, and the $\sqrt{N}$ Gaussian fluctuation limit is unchanged. No globally bounded kinetic likelihood is asserted.

With this reference for every copy and the same β_N in (5.3), the phase events of Section 5.4 keep their conditional limits by Lemma 5.4. Theorem 5.1 then gives canonical one-copy marginals and joint TV error 1/2, and Proposition 5.2 gives one bit of mutual information. The law reproduced is the balanced canonical law at β_N , not the unbalanced law at β_c .

6 Smooth reservoirs and the boundary law

This section extends the two-phase criteria of Section 3 to smooth reservoir entropies and computes the exact-reservoir error at $c_N \sim \gamma N^{3/2}$, where each phase acquires a finite Gaussian tilt.

6.1 Smooth reservoir entropies

Let a reservoir induce a log likelihood $h_N(E)$ relative to a specified canonical law. For reservoir entropy $s_{B,N} = S_{B,N}/k_B$, $h_N(E) = \beta_N E + s_{B,N}(\mathcal{E}_N - E)$ up to a constant. Where temperature and positive heat capacity are defined by this entropy convention, the curvature of h_N is a heat-capacity ratio:

$$-h_N''(E) = -s_{B,N}''(\mathcal{E}_N - E) = \frac{[\beta_B(\mathcal{E}_N - E)]^2}{C_B(\mathcal{E}_N - E)/k_B}. \quad (6.1)$$

Proposition 6.1 (Curvature necessity). *Suppose a positive decomposition (3.9) has two phases with positive limiting weights, centers $e_{-,N} < e_{+,N}$ separated by $\Delta_N \asymp N$, and conditional limits $(E - e_{i,N})/\sqrt{N} \Rightarrow \mathcal{N}(0, v_i)$ with $v_i > 0$. Let Q_N have likelihood proportional to e^{h_N} , with $h_N = -\infty$ outside an interval. On its interior assume h_N is twice continuously differentiable and $-h_N'' \geq k\kappa_N$ between the phase centers, for some $k > 0$ and a positive sequence $\kappa_N > 0$. Then $d_{TV}(P_N, Q_N) \rightarrow 0$ requires $\kappa_N N^{3/2} \rightarrow 0$.*

Proof. We adapt the good-point chord argument for Theorem 3.4. TV convergence gives $h_N(E) - \log Z_N \rightarrow 0$ in canonical probability on the feasible set. Choose good energies with standardized positions in $[1/2, 1]$ and $[3/2, 2]$ above the lower center, and $[-1, -1/2]$ below the upper center, each with positive limiting probability. The selected feasible energies $x_N < y_N < z_N$ have log likelihoods within $o(1)$ of $\log Z_N$, $y_N - x_N \asymp \sqrt{N}$, $z_N - y_N \asymp N$, and a feasible convex hull between the centers. Concavity of $h_N(E) + k\kappa_N E^2/2$ gives

$$h_N(y_N) - \frac{z_N - y_N}{z_N - x_N} h_N(x_N) - \frac{y_N - x_N}{z_N - x_N} h_N(z_N) \geq \frac{k\kappa_N}{2} (y_N - x_N)(z_N - y_N).$$

The left side tends to zero; the right side is of order $\kappa_N N^{3/2}$. □

Under the density hypotheses of Section 3.4, Proposition 3.7 gives the converse when $h_N(e_{-,N}) = h_N(e_{+,N}) = 0$, h_N is globally concave on its feasible interval, and its curvature satisfies (3.19). Hence

$\kappa_N N^{3/2} \rightarrow 0$ is sharp for curvature comparable to κ_N , provided a composite energy equalizes h_N at the two centers, which for an arbitrary entropy need not exist. Exactly Gaussian separated phases need no interphase density hypothesis: the chord and curvature upper bound make the standardized endpoint slopes vanish, and Gaussian exponential moments make the tangent bound uniformly integrable, giving the analogous criterion $\kappa_N \Delta_N s_N \rightarrow 0$ for gap Δ_N and Gaussian fluctuation scale s_N .

6.2 Finite tilts at capacity proportional to $N^{3/2}$

For the exact reservoir, use the positive decomposition (3.9) with $w_{i,N} \rightarrow w_i > 0$, $w_- + w_+ = 1$, and $\Delta_N/N \rightarrow \ell > 0$. Write $\mathcal{E}_N^{\text{sec}}$ and h_N^{sec} for the secant calibration (2.10) and its log likelihood (2.11). A Gaussian $G_i = \mathcal{N}(0, v_i)$ with $v_i = 0$ is a point mass.

Theorem 6.2 (Boundary law and finite calibration corrections). *Suppose $c_N/N^{3/2} \rightarrow \gamma \in (0, \infty)$ and*

$$\begin{aligned} X_{i,N} &:= (E_N - e_{i,N})/\sqrt{N} \Rightarrow G_i \text{ under } P_{i,N}, \\ \sup_{N \geq N_0} \mathbb{E}_{P_{i,N}} e^{\eta |X_{i,N}|} &< \infty \quad \text{for some } N_0 \text{ and } \eta > b := \frac{\beta^2 \ell}{2\gamma}. \end{aligned} \quad (6.2)$$

Assume also

$$\delta_N \exp \left[\max_E h_N^{\text{sec}}(E) \right] \rightarrow 0. \quad (6.3)$$

For $\mathcal{E}_N = \mathcal{E}_N^{\text{sec}} + \xi_N \sqrt{N}$, $\xi_N \rightarrow \xi$, put $b_- = b$, $b_+ = -b$, $D = \beta^2 \ell \xi / \gamma$, and

$$A_- = 0, \quad A_+ = D, \quad Z(\xi) = \sum_i w_i e^{A_i + b^2 v_i / 2}, \quad r_i(\xi) = \frac{w_i e^{A_i + b^2 v_i / 2}}{Z(\xi)}. \quad (6.4)$$

The reservoir phase weights, defined through the positive decomposition (3.9) (possibly on an enlarged probability space), tend to $r_i(\xi)$; their standardized conditional fluctuations converge weakly to $G_i^b := \mathcal{N}(b_i v_i, v_i)$. Because the phases are separated on scale N , the same limits hold for the reservoir probabilities of the energy events below and above the midpoint of the centers. On the full microscopic state space,

$$\lim_N d_{\text{TV}}(P_N, Q_N) = \frac{1}{2} \sum_i w_i \int_{\mathbb{R}} \left| \frac{e^{A_i + b_i z}}{Z(\xi)} - 1 \right| dG_i(z). \quad (6.5)$$

Proof. Normalize h_N at $e_{-,N}$. Uniformly on bounded standardized windows, the exact secant formula gives

$$h_N(e_{i,N} + \sqrt{N}z) = A_i + b_i z + o(1). \quad (6.6)$$

Indeed, with $q_N = \beta_N \Delta_N / c_N \rightarrow 0$, the secant endpoint derivatives are $\beta_N q_N / 2 + O(q_N^2)$ and $-\beta_N q_N / 2 + O(q_N^2)$; their products with $\sqrt{N}$ tend to b and $-b$, and $N \sup |h_N''| = O(N/c_N) \rightarrow 0$ on these windows. Changing $\mathcal{E}_N$ by $\xi_N \sqrt{N}$ changes the upper-minus-lower endpoint value by

$$\xi_N \sqrt{N} c_N \left[(\mathcal{E}_N^{\text{sec}} - e_{+,N})^{-1} - (\mathcal{E}_N^{\text{sec}} - e_{-,N})^{-1} \right] + o(1) \rightarrow D,$$

and each standardized endpoint slope by $o(1)$.

For the tails, concavity gives $h_N(E) \leq h_N(e_{i,N}) + h_N'(e_{i,N})(E - e_{i,N})$ on the feasible set, so $e^{h_N(E_N)} \leq C \exp[(b + \epsilon)|X_{i,N}|]$ for small fixed $\epsilon > 0$ and large N . Choosing $p > 1$ with $p(b + \epsilon) < \eta$, (6.2) gives uniform integrability, also for the absolute difference in (6.5); energies at or above the

cutoff have zero weight. For bounded ξ_N , the maximum of h_N , at $E = \mathcal{E}_N - c_N/\beta_N$, differs from the secant maximum by $O(1)$: when $h_N(e_{-,N}) = 0$, the derivative of its value with respect to $\mathcal{E}_N$ is $\beta_N - c_N/(\mathcal{E}_N - e_{-,N}) = O(N^{-1/2})$ in this range. Thus (6.3) removes the exceptional contribution. Weak convergence, (6.6), and uniform integrability give $Z_N \rightarrow Z(\xi)$ and the Gaussian tilts and, with the exact identity $d_{\text{TV}}(P_N, Q_N) = \frac{1}{2} \mathbb{E}_{P_N} |e^{h_N}/Z_N - 1|$, also (6.5). Finally, the same uniform integrability and tightness on scale $\sqrt{N}$ show that each reweighted phase puts vanishing reservoir mass across the midpoint of the centers, and the exceptional contribution vanishes; this gives the limits for the energy events below and above the midpoint. $\square$

The secant calibration $\xi = 0$ equalizes the likelihoods at the centers; it preserves the phase weights ($r_i(0) = w_i$) exactly when $v_- = v_+$. For $v_- = v_+ = v$,

$$\lim_N d_{\text{TV}}(P_N, Q_N) = 2\Phi(b\sqrt{v}/2) - 1, \quad (6.7)$$

where Φ is the standard normal distribution function.

Corollary 6.3 (Compensating the phase weights). *Under these hypotheses, the calibration*

$$\mathcal{E}_N = \mathcal{E}_N^{\text{sec}} + \frac{\beta^2 \ell(v_- - v_+)}{8\gamma} \sqrt{N} + o(\sqrt{N}) \quad (6.8)$$

restores the original phase weights asymptotically, and

$$\lim_N d_{\text{TV}}(P_N, Q_N) = \sum_i w_i \{2\Phi(b\sqrt{v_i}/2) - 1\}. \quad (6.9)$$

Proof. The correction makes $D = b^2(v_- - v_+)/2$, so the integrated factors in (6.4) are equal. Within each phase the remaining change is the Gaussian translation by $b_i v_i$, with TV $2\Phi(b\sqrt{v_i}/2) - 1$. This is zero when $v_i = 0$. $\square$

Restoring the phase weights thus leaves a finite distortion in each phase with $v_i > 0$.

6.3 Optimization over all calibrations

To optimize over all calibrations, not only shifts of order $\sqrt{N}$, let $\|\cdot\|_{\text{var}}$ be the total variation norm of finite signed measures and define the continuous convex function

$$F(r) = \frac{1}{2} \left\{ \|w_- G_- - r G_-^b\|_{\text{var}} + \|w_+ G_+ - (1-r) G_+^b\|_{\text{var}} \right\}, \quad 0 \leq r \leq 1. \quad (6.10)$$

Theorem 6.4 (Optimized boundary). *Under Theorem 6.2's hypotheses,*

$$\lim_N \inf_{\mathcal{E}: Z_N(\mathcal{E}) > 0} d_{\text{TV}}(P_N, Q_{N,\mathcal{E}}) = \min \left\{ \min_{0 \leq r \leq 1} F(r), w_-, w_+ \right\}. \quad (6.11)$$

For $w_- = w_+ = 1/2$ and $v_- = v_+ = v$, this is $\min\{2\Phi(b\sqrt{v}/2) - 1, 1/2\}$.

Proof. For the upper bound, as ξ varies through $\mathbb{R}$, (6.4) realizes every strictly positive pair of phase weights, so the boundary theorem realizes $F(r)$ for every $r \in (0, 1)$; continuity extends this to the infimum on $[0, 1]$. Centering the reservoir maximum at either phase, as in (2.7), gives the upper bounds w_- and w_+ : its weight lies in $[0, 1]$, tends to one in the selected phase since $N/c_N \rightarrow 0$, and to zero in the other since $\Delta_N^2/c_N \rightarrow \infty$, so positive decomposition and macroscopic separation give TV tending to the discarded phase weight.

For the lower bound consider a subsequence of arbitrary calibrations with TV limit strictly below $\min(w_-, w_+)$, and let R_N be their normalized likelihood. Since $d_{\text{TV}}(P_N, Q_N) = \mathbb{E}_{P_N}(1 - R_N)_+$, each phase loss is at most TV, so the overlap $O_{i,N} = w_{i,N} \mathbb{E}_{P_{i,N}} \min(R_N, 1) \geq w_{i,N} - d_{\text{TV}}(P_N, Q_N)$ stays positive in both phases. Tightness, the overlap bound from $R_N < \epsilon$, and $P_N(R_N > M) \leq 1/M$ allow selection of feasible points $x_N = e_{-,N} + O(\sqrt{N})$ and $y_N = e_{+,N} + O(\sqrt{N})$ with $R_N(x_N), R_N(y_N) \in [\epsilon, M]$, for fixed positive ϵ, M . Set $\zeta_N = \log R_N(y_N) - \log R_N(x_N) = O(1)$ and $D_N = y_N - x_N \asymp N$. Solving the exact likelihood for the composite energy gives

$$\mathcal{E}_N = x_N + \frac{D_N}{1 - \exp[-(\beta_N D_N - \zeta_N)/c_N]}. \quad (6.12)$$

A bounded ζ_N changes this by $O(c_N/D_N) = O(\sqrt{N})$ from the secant calibration for x_N, y_N , which uniformly on these windows is

$$\frac{c_N}{\beta_N} + \frac{x_N + y_N}{2} + \frac{\beta_N(y_N - x_N)^2}{12c_N} + O\left(\frac{(y_N - x_N)^4}{c_N^3}\right).$$

Moving its endpoints by $O(\sqrt{N})$ changes it by $O(\sqrt{N})$, so $(\mathcal{E}_N - \mathcal{E}_N^{\text{sec}})/\sqrt{N}$ is bounded. On a convergent subsequence, Theorem 6.2 gives the limit $F(r) \geq \min F$. Sequences with limiting error at least $\min(w_-, w_+)$ already obey the bound. Applied to calibrations within $1/N$ of the infimum (no minimizer need exist), this makes the upper and lower bounds coincide, so the limit in (6.11) exists and has the stated value. In the balanced equal-variance case reflection sends r to $1 - r$, so convexity minimizes F at $1/2$, giving the normal formula. $\square$

To compute $\min F$, set $r_- = r$, $r_+ = 1 - r$, and $\lambda_i = b\sqrt{v_i}$. For $\lambda_i > 0$ the contribution of phase i to the positive part of reservoir minus reference, $r_i G_i^b - w_i G_i$, is

$$r_i \Phi\left(\frac{\log(r_i/w_i)}{\lambda_i} + \frac{\lambda_i}{2}\right) - w_i \Phi\left(\frac{\log(r_i/w_i)}{\lambda_i} - \frac{\lambda_i}{2}\right). \quad (6.13)$$

Equation (6.13) follows by integrating over the half-line where the tilted Gaussian density exceeds the weighted original density. Use zero for $r_i = 0$, by continuity, and $(r_i - w_i)_+$ when $\lambda_i = 0$. The two contributions sum to $F(r)$ because the positive and negative masses agree. If both $\lambda_i > 0$, differentiation gives

$$F'(r) = \Phi\left(\frac{\log(r/w_-)}{\lambda_-} + \frac{\lambda_-}{2}\right) - \Phi\left(\frac{\log((1-r)/w_+)}{\lambda_+} + \frac{\lambda_+}{2}\right).$$

This is strictly increasing from a negative to a positive value, so its unique zero, where the two CDF arguments are equal, is the unique minimizer. For equal positive variances this is $r = w_-$, even with unequal target weights, and the optimized limit is $\min\{2\Phi(b\sqrt{v}/2) - 1, w_-, w_+\}$.

6.4 The exact interior gain of the secant calibration

Hypothesis (6.3) involves the maximum of h_N^{sec} , which is explicit for the exact reservoir: at a fraction θ of the gap,

$$h_N(e_{-,N} + \theta \Delta_N) = c_N [\theta q_N + \log(1 - \theta + \theta e^{-q_N})], \quad q_N = \frac{\beta_N \Delta_N}{c_N}. \quad (6.14)$$

For $q_N \rightarrow 0$, expanding the logarithm of the Bernoulli transform in (6.14) gives, uniformly on the bounded range $0 \leq \theta \leq 1$,

$$\begin{aligned} h_N(e_{-,N} + \theta \Delta_N) &= \frac{\beta_N^2 \Delta_N^2}{2c_N} \theta(1 - \theta) \\ &\quad + \frac{\beta_N^3 \Delta_N^3}{6c_N^2} \theta(1 - \theta)(2\theta - 1) + O(\Delta_N^4/c_N^3). \end{aligned} \quad (6.15)$$

The maximum lies between the centers since the endpoint slopes have opposite signs. When $c_N/N^{3/2} \rightarrow \gamma$ and $\Delta_N/N \rightarrow \ell$, the quadratic term is maximized at $\theta = 1/2$, and the other terms, including the cubic one, are $O(1)$. They do not change the $\sqrt{N}$ limit, but the quadratic term alone does not determine the $O(1)$ weights. Hence (6.15) gives

$$\frac{\max_E h_N^{\text{sec}}(E)}{\sqrt{N}} \rightarrow \frac{\beta^2 \ell^2}{8\gamma}. \quad (6.16)$$

6.5 Microscopic applications of the boundary law

We verify the hypotheses of Theorems 6.2 and 6.4 for the models of Section 4.

For the mean-field model of Theorem 4.1, the boundary law holds for every $\gamma > 0$ and kinetic parameter $a \geq 0$. The occupation bound of Lemma 4.2 is Gaussian on the $\sqrt{N}$ scale, so it bounds $\mathbb{E}e^{t|X_{i,N}|}$ for every fixed t and large N ; so does the independent Gamma contribution, by its exact transform, and there is no exceptional component. The variances are (4.10); pure spins ($v_+ = 0$) are allowed, so no kinetic energy is needed.

For the short-range model of Theorem 4.3, the boundary law and its optimization hold for every $\gamma > 0$ in fixed spatial dimension $d \geq 3$, and for all sufficiently large fixed γ in $d = 2$. Since Lemma A.2 gives only a small fixed exponential moment, we use the cutoff-free derivative bounds in the window of width η_0/L , where η_0 is the constant η of Lemma A.1. For $d > 2$ every fixed multiple of $N^{-1/2} = L^{-d/2}$ eventually lies in that window, so the second-derivative bound (A.15) and center estimate (A.16) bound exponential moments of the standardized bond count at every fixed parameter. The change from inverse temperature to bond fugacity has bounded derivatives, and the conditional binomial estimate and Cauchy–Schwarz argument of Lemma A.2 transfer these bounds to the spin energy. The weak limits for the contour phases agree with (A.13), proved for the midpoint phases: the small exponential moment bounds the conditional probability of the opposite macroscopic midpoint phase by $Ce^{-c\sqrt{N}}$, the exceptional contour probability vanishes, and positive limiting phase weights identify the conditional laws.

The exceptional contour probability is at most $Ce^{-\tau_* L^{d-1}}$. For $d > 2$ this surface exponent dominates the gain (6.16) of order $\sqrt{N} = L^{d/2}$, giving (6.3) for every γ . For $d = 2$, with $\eta_* > 0$ a valid spin-phase moment coefficient and $\tau_* > 0$ a valid exceptional surface constant, it suffices that

$$\frac{\beta^2 \ell}{2\gamma} < \eta_*, \quad \frac{\beta^2 \ell^2}{8\gamma} < \tau_*. \quad (6.17)$$

Both conditions hold for all sufficiently large γ ; for smaller γ in $d = 2$ these estimates assert nothing.

Instead of (6.2) and (6.3), one may assume the continuous density envelope (3.18) with $\alpha > 1/2$ and (3.17); these also give the boundary law and its finite corrections for every fixed γ . Put $x = \min_i |E - e_{i,N}|$ on the interior. At the scaling $c_N \sim \gamma N^{3/2}$, concavity gives $h_N^{\text{sec}}(E) \leq Cx/\sqrt{N}$; with a finite calibration correction the endpoint values are bounded but need not both vanish, so $h_N(E) \leq C_0 + C_1 x/\sqrt{N}$ with constants independent of N . For $x \geq R\sqrt{N}$ the ratio of the linear term to x^2/N is at most C_1/R , and its ratio to x^α is at most $C_1 N^{1/2-\alpha}$, so choosing R and then N

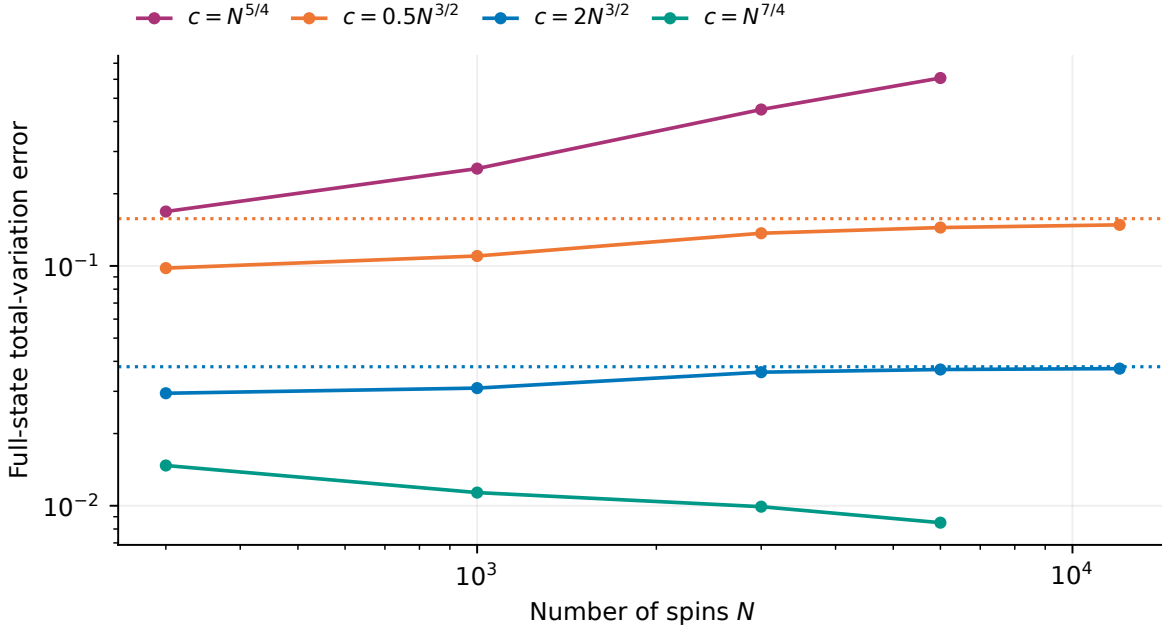


Figure 1: Exact finite-system mean-field Potts calculations with the exact reservoir at the secant calibration. Points are occupation sums and Gamma/Beta CDF evaluations, connected to guide the eye. Dotted lines are the proved secant boundary limits for $c_N = \gamma N^{3/2}$, with $\gamma = 0.5$ and 2 . Only these two sequences extend to $N = 12000$; the others stop at $N = 6000$. These are mean-field data and do not test a short-range interfacial law.

large absorbs this term in the envelope; the offset adds only the bounded prefactor e^{C_0} to the tail integral, whose limit as $R \rightarrow \infty$ still vanishes. Outside the centers the secant weight is at most one, and with finite corrections concavity and the endpoint slope signs bound the exterior weights by a constant, which suffices. Weighted tail control and local density convergence then give the same integral formula. At $\alpha = 1/2$ the second ratio need not vanish.

Both $\alpha = 1/2$ and the restricted range (6.17) have a scale explanation: an interface with surface cost of order $L^{d-1} = N^{(d-1)/d}$ competes with a maximal reservoir gain of order N^2/c_N , and they balance at $c_N \asymp N^{1+1/d}$, below the $N^{3/2}$ fluctuation threshold for $d > 2$ and equal to it for $d = 2$. A surface bound alone does not identify a favored droplet or slab; Appendix C analyzes a stated capillarity model for this comparison.

7 Deterministic numerical illustrations

Figure 1 shows exact finite- N calculations for the mean-field model with $J = 1$, $\beta = 4 \log 2$, and $a = 1/2$. Occupation triples are enumerated modulo color permutations, and the Gamma and Beta conditional energy integrals are evaluated through their distribution functions (Section 4), so the plotted quantity is the full-state TV of the spin-momentum law. No Monte Carlo sampling or Gaussian approximation enters the finite- N points.

At $c_N = 0.5N^{3/2}$ the predicted limit is approximately 0.157392 and the exact $N = 12000$ value is 0.148341; at $c_N = 2N^{3/2}$ they are 0.037986 and 0.037304. These are errors at the secant calibration, not finite- N optima. The figure also shows $c_N = N^{7/4}$, above the threshold, where the error decays, and $c_N = N^{5/4}$, below it, where it persists. Such finite sizes cannot establish an asymptotic exponent;

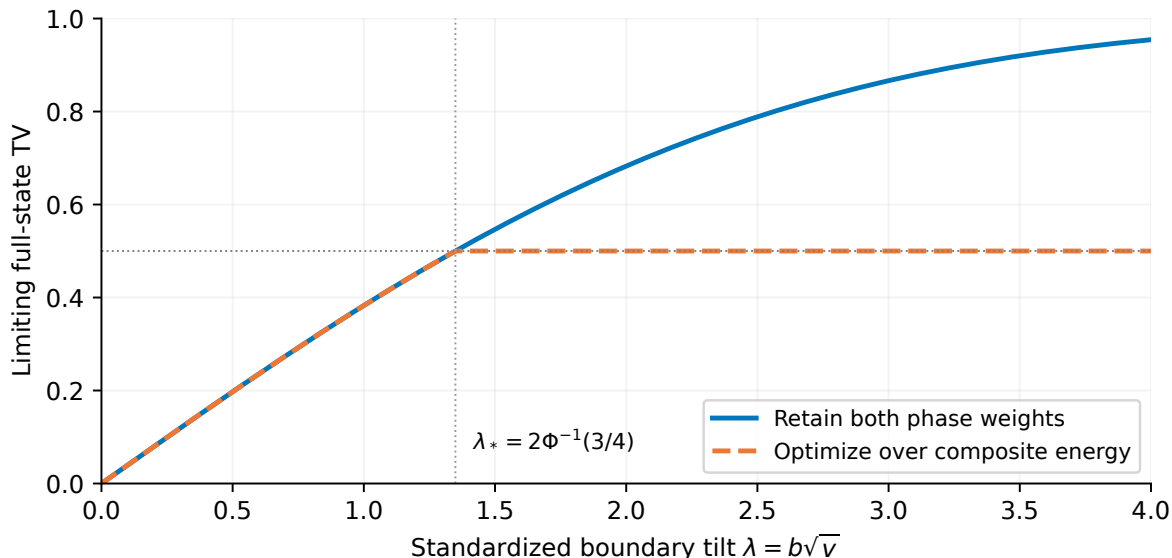


Figure 2: The boundary law for balanced equal-variance phases under Theorem 6.2’s moment and exceptional-mass hypotheses, evaluated from the limiting formula (not finite-size Potts data). At $\lambda_* = 2\Phi^{-1}(3/4) \simeq 1.34898$, the best calibration that retains both phases and the one that discards a phase have the same limiting error. The same curve holds for the exact scalar Gaussian model of Proposition B.1.

the theorems do.

These sizes are also far from the asymptotic regime of the phase decomposition. The saddle of the rate function (4.3) at $(1/2, 1/4, 1/4)$ lies on the Voronoi boundary between the ordered and disordered cells, with rate excess $I(1/2, 1/4, 1/4) - I_{\min} = \log 3 - \frac{19}{12} \log 2 \simeq 1.1 \times 10^{-3}$, so the interphase valley is suppressed only by $\exp(-0.0011N)$, of order one below $N \simeq 10^3$. Exact pure-spin occupation sums show the slow approach. With Euclidean Voronoi distance in the three occupation fractions and ties assigned to the ordered phase, so that the combined ordered event is $\max_j n_j \geq N/2$, the Voronoi-conditional ordered variance divided by N is 1.02 at $N = 3000$ and 0.78 at $N = 12000$, against the pure-spin limit $v_- = J^2/[6(3 - \beta J)] \simeq 0.733$ (the $a = 0$ case of (4.10)), while the ordered weight is 0.528 and 0.529 against $w_- \simeq 0.530$ from (4.9). The dotted lines use the limiting constants.

Figure 2 shows the proved limiting formula for balanced phases of equal variance, not finite-size data. Writing $\lambda = b\sqrt{v}$, preserving the phase weights gives $2\Phi(\lambda/2) - 1$, whereas optimizing the calibration gives $\min\{2\Phi(\lambda/2) - 1, 1/2\}$, whose second branch discards one phase.

The accompanying source bundle contains the finite-system data, figure code, and independent checks. Small systems are checked against explicit enumeration of all 3^N labeled spin states for $1 \leq N \leq 8$; at $N = 12$, direct absolute-density integration verifies the Gamma/Beta crossing formula for two kinetic shapes; and separate normal-density quadratures verify the weighted CDF expression (6.13) and its optimizer. These floating-point calculations are numerical checks, not interval certificates or substitutes for the proofs. The reproducibility instructions separate quick checks and figure regeneration from the expensive full occupation enumeration through $N = 12000$.

8 Discussion and conclusions

A reservoir requirement is precise only once we specify what must be reproduced. The results separate three ingredients that are often merged: the reservoir capacity, its calibration, and the observable being tested. For the complete canonical law at coexistence, capacity and calibration interact: the secant calibration removes the phase bias caused by the latent heat, so the capacity needs to control only the tilt within each phase. This is why two-phase coexistence needs $c_N \gg N^{3/2}$ rather than the N^2 suggested by the latent heat or by the subsystem's canonical heat capacity. Three distinct phase energies, including the combined energy of two coexisting copies, restore the N^2 requirement. Smooth reservoirs with positive heat capacity obey the same curvature mechanism when a suitable calibration and tail control are available. The heat-capacity convention also matters at finite size, as finite ideal-gas contact calculations show [11].

The observable matters as much. Phase weights should be reported with any optimized error, since at the threshold scale the smallest full-state TV can come from discarding a phase. Marginal and joint accuracy differ: with balanced phase weights, a shared reservoir can leave each copy canonical while anticorrelating their phases, an equilibrium effect that needs no direct interaction between the copies. Other criteria are not equivalent to full-state accuracy: a small Euclidean histogram norm or macroscopic energy concentration does not imply it, and the log-density barrier between a phase center and the interphase midpoint can remain lowered by a large amount while the full-state TV tends to zero.

The proved scope is equilibrium coupling to the stated reservoir with additive energies, fixed positive limiting phase weights, and the explicit fluctuation and tail hypotheses of each theorem. The short-range results require a fixed number of colors $q \geq q_0(d)$. Their threshold-scale law holds for every fixed capacity coefficient $\gamma > 0$ in dimensions above two, and for sufficiently large γ in two dimensions. The square-torus interfacial calculation is a stated capillarity model, not a derivation of microscopic droplet or slab morphology, and the Gaussian, capillarity, and linear-capacity appendices analyze stated models exactly, whereas the contour appendix proves a microscopic hypothesis.

These limits suggest open questions: the two-dimensional threshold-scale law for small γ , where interfacial configurations compete with the phases; Potts models with fewer colors and other interacting systems; interacting spatial subregions of a single system, which a shared-reservoir model of independent copies does not describe; quantum systems, in particular observables that do not commute with the energy; and explicit finite-size error bounds.

A Positive contour measures and the short-range theorem

This appendix proves Theorem 4.3. Section A.1 collects the published inputs and fixes the energy normalization. Section A.2 proves weak Gaussian limits with positive variances for the midpoint energy events, which gives necessity. Section A.3 bounds the second derivatives of the logarithms of the positive restricted contour partition functions in a window of width $O(1/L)$ around β_c , and Section A.4 transfers the resulting occupied-bond moment bounds to the spin energy and completes the proof.

The color threshold $q_0(d)$ is fixed here as the largest of: the thresholds required by the cited finite-size scaling and contour results [4, 3] on a fixed temperature interval around coexistence; the value $2d + 1$, used for the variance lower bound in (A.14); and the value needed for the cluster-expansion criterion of [3] to hold with the extra factor $e^{\rho m \gamma}$ in (A.19) for a fixed $\rho > 0$. Every “sufficiently large q ” below refers to this threshold.

A.1 Published finite-volume inputs and normalization

Theorem 1 of Borgs, Kotecký, and Miracle-Solé [4] (BKMS) gives six-times differentiable dimensionless free energies ψ_o, ψ_d and a fixed real interval containing β_c on which

$$Z_N^{\text{spin}}(\beta) = qe^{-N\psi_o(\beta)} + e^{-N\psi_d(\beta)} + O(e^{-bL})e^{-N \min_i \psi_i(\beta)}. \quad (\text{A.1})$$

The constants are uniform on a sufficiently small fixed compact interval; the source’s error $O(q^{-bL})$ is $O(e^{-bL})$ for fixed q , with a different b . At coexistence the free energies are equal and

$$u_i = \psi'_i(\beta_c), \quad u_o < u_d, \quad \sigma_i^2 = -\psi''_i(\beta_c). \quad (\text{A.2})$$

The ordered branch is stable above β_c , and the disordered branch below. The source Hamiltonian, its equation (6) with $J = 1$, is (4.19), and our ψ_i is its βf_i . No complex-temperature continuation of (A.1) is assumed.

The positive decomposition lives on bond sets $A \subseteq \mathcal{B}_L$. With $p = 1 - e^{-\beta}$, their random-cluster weights are

$$w_\beta(A) = p^{|A|}(1-p)^{dN-|A|}q^{k(A)}, \quad Z_N^{\text{RC}}(\beta) = \sum_A w_\beta(A) = e^{-\beta dN} Z_N^{\text{spin}}(\beta). \quad (\text{A.3})$$

This normalization, obtained by expanding $\prod_{\{x,y\}}[1 + (e^\beta - 1)\mathbf{1}_{\{\sigma_x = \sigma_y\}}]$, fixes the additive energy convention independently of the contour references.

Throughout, contours, interiors and exteriors, compatibility, phase events, restricted partition functions, and activities are those of Borgs, Chayes, and Tetali [3] (BCT), Definitions 4.1–4.4 and 5.4, equations (6.1)–(6.5) and (6.13)–(6.22). BCT’s exterior of a contour is the component containing a compatible interface if one exists, and otherwise the larger component, with a fixed tie rule; this matters for large contours on the torus. Following BCT, γ denotes a contour in this appendix; it is unrelated to the capacity coefficient γ of Section 6.

The construction thickens the occupied bonds and separates ordered and disordered regions by their boundary components: components with zero winding vector are contours, and the others are interfaces. Without interfaces, the common exterior of the contours has one label, ordered or disordered, which defines the positive event $\mathcal{O}_L$ or $\mathcal{D}_L$; configurations with an interface form the tunneling event $\mathcal{T}_L$. These events partition the bond configurations and do not depend on β . With $Z_{o,L}$ the ordered contribution (color multiplicity removed), $Z_{d,L}$ the disordered one, and $Z_{t,L}$ the tunneling one, BCT (6.1)–(6.5) give exactly

$$Z_N^{\text{RC}} = qZ_{o,L} + Z_{d,L} + Z_{t,L}. \quad (\text{A.4})$$

At β_c , BCT Lemma 6.1 bounds the tunneling probability by $Ce^{-bL^{d-1}}$. The later construction of [2], Definition 5, uses a modified exterior convention; we import nothing from it.

Write $m_\gamma = \|\gamma\|$ for the size of a contour. Its diameter and interior obey

$$\text{diam } \gamma \leq L, \quad \text{diam } \gamma \leq m_\gamma/2, \quad |\text{Int } \gamma \cap \Lambda_L| \leq \frac{1}{2}m_\gamma \text{diam } \gamma. \quad (\text{A.5})$$

These follow from the definition of diameter and Lemma 5.7 of [3], for all $d \geq 2$. The positive restricted partition functions have the polymer form

$$Z_{i,L} = e^{-e_i(\beta)N} \sum_{\Gamma \text{ compatible}} \prod_{\gamma \in \Gamma} K_i(\gamma; \beta). \quad (\text{A.6})$$

The ground coefficients are $e_o = -d \log(1 - e^{-\beta})$ and $e_d = d\beta - \log q$. With $\kappa = \frac{1}{2} \log(e^\beta - 1)$, the activities are $e^{-\kappa(\beta)m_\gamma}$ times a ratio of positive interior partition functions, with a factor q in the disordered case (BCT (6.18)–(6.22)). The coefficients e_i and κ have bounded derivatives on the fixed temperature interval.

BCT Appendix A defines truncated activities

$$K'_i(\gamma; \beta) = \min\{K_i(\gamma; \beta), e^{-\tau(\beta)m_\gamma}\}, \quad (\text{A.7})$$

where BCT (A.3) uses $\tau(\beta) = \beta/8 - \beta/20 + 1$; this cap ensures uniform convergence of the cluster expansion for large q . Write $Z'_{i,L}$ for the truncated partition functions and f_i^{tr} for their infinite-volume free energies (distinct from the ψ_i in (A.1)). If $a_i = f_i^{\text{tr}} - \min_j f_j^{\text{tr}}$, BCT Lemma A.1(i) states

$$a_i(\beta) \text{diam } \gamma \leq c_0 \beta \implies K_i(\gamma; \beta) = K'_i(\gamma; \beta). \quad (\text{A.8})$$

Moreover, BCT (A.8) on the full torus gives

$$|\log Z'_{i,L}(\beta) + Nf_i^{\text{tr}}(\beta)| \leq CNe^{-bL}. \quad (\text{A.9})$$

The assumptions of BCT Appendix A cover a fixed neighborhood on both sides of β_c , although BCT's Lemma 6.3 is stated on the ordered side.

It remains to identify the stable sides and the truncated free energies with the data in (A.1). BCT Lemma A.3 says that the spontaneous magnetization is positive exactly when $a_o = 0$. BCT define their transition point β_0 as the onset of positive magnetization, whereas β_c above is the crossing $\psi_o = \psi_d$ of the BKMS free energies. The two coincide: for $q \geq q_0(d)$ the model has a unique, disordered Gibbs state below the large- q transition point and q ordered extremal states above it [23, 24], and the BKMS expansion (A.1) is derived from the same Pirogov–Sinai construction, in which the phase with smaller ψ_i is stable at each β . Hence the magnetization vanishes below β_c and is positive above it, so $a_d = 0$ below and $a_o = 0$ above. Continuity gives $f_o^{\text{tr}} = f_d^{\text{tr}}$ at coexistence.

The minimum $f^{\text{tr}} = \min_i f_i^{\text{tr}}$ is the physical thermodynamic random-cluster free energy. For the untruncated partition functions on the full torus, BCT Lemma A.1(ii) gives $Z_{i,L} \leq \exp[-Nf^{\text{tr}} + O(Ne^{-bL})]$, while BCT (A.9) supplies the corresponding lower bound for any minimizing phase. The interface-network sum BCT (6.26)–(6.27) bounds $Z_{t,L}$ by $\exp[-Nf^{\text{tr}} - bL^{d-1}]$. That sum uses only the bounds of BCT Appendix A and the large- q contour counting inequalities, valid throughout the fixed neighborhood (shrunk if necessary so that $q \leq e^{2d\kappa+d}$ there, as BCT (6.27) requires), and does not require the ordered phase to minimize. With (A.4) these estimates identify the limiting pressure, and (A.3) with the stable branch of (A.1) gives

$$f_i^{\text{tr}}(\beta) = \psi_i(\beta) + d\beta \quad \text{on the stable side of phase } i. \quad (\text{A.10})$$

A.2 Weak phase energy limits and nonzero variances

Applying (A.1) at $\beta_c + t/N$ and using the boundedness of U_N/N gives

$$U_N/N \Rightarrow \frac{q}{q+1}\delta_{u_o} + \frac{1}{q+1}\delta_{u_d}. \quad (\text{A.11})$$

The partition-function ratios are Laplace transforms that converge for every fixed real t to those of this bounded law. The midpoint energy events $B_{o,N} = \{U_N/N < (u_o + u_d)/2\}$ and $B_{d,N} = B_{o,N}^c$ therefore have the limiting probabilities in (A.11).

For $s > 0$, use (A.1) at $\beta_c + s/\sqrt{N}$. The ordered branch dominates, and its quadratic Taylor expansion gives

$$\mathbb{E}_{\beta_c} \exp[-s(U_N - Nu_o)/\sqrt{N}] \longrightarrow \frac{q}{q+1} e^{\sigma_o^2 s^2/2}. \quad (\text{A.12})$$

The part on $B_{d,N}$ is at most $e^{-s(u_d - u_o)\sqrt{N}/2}$ and vanishes, so this is also the transform limit of the positive subprobability measure restricted to $B_{o,N}$. The disordered phase uses the opposite sign and $\beta_c - s/\sqrt{N}$.

This one-sided limit suffices for weak convergence. Tilt the ordered subprobability law of $x = (U_N - Nu_o)/\sqrt{N}$ by $e^{-s_0 x}$ for any fixed $s_0 > 0$ and normalize it. Ratios of (A.12) at $s_0 - t$ and s_0 , for $|t| < s_0$, give convergence of its moment-generating function to that of $\mathcal{N}(-\sigma_o^2 s_0, \sigma_o^2)$, so the tilted probabilities converge weakly by the moment-generating-function convergence theorem. Untilting against compactly supported continuous functions gives vague convergence of the original measures; since their total masses converge to the recovered Gaussian mass, the convergence is weak. Hence

$$(U_N - Nu_i)/\sqrt{N} \mid B_{i,N} \Rightarrow \mathcal{N}(0, \sigma_i^2), \quad i = o, d. \quad (\text{A.13})$$

A priori $\sigma_i^2 \geq 0$, by convexity of the limiting log transforms.

In fact both variances are strictly positive. Choose an independent vertex set S of size at least $N/(2d+1)$ and condition on all spins outside it, which makes the spins in S independent. For $x \in S$, let $n_{x,a}$ count the neighbors of color a . Then $\sum_a n_{x,a} = 2d$, and the conditional color probabilities on a compact inverse-temperature interval with upper endpoint $\beta_{\max}$ obey $p_{x,a} \geq q^{-1}e^{-2d\beta_{\max}}$. If $q > 2d$, two neighbor counts differ by at least one. Since $\text{Var}_p(n) = \sum_{a < b} p_a p_b (n_a - n_b)^2$, conditional variance gives

$$\text{Var}_\beta U_N \geq v_* N, \quad v_* := \frac{e^{-4d\beta_{\max}}}{(2d+1)q^2} > 0. \quad (\text{A.14})$$

Thus $N^{-1} \log Z_N^{\text{spin}} - v_* \beta^2/2$ is convex, and so is its pointwise thermodynamic limit. On each open stable side that limit equals $-\psi_i(\beta) - v_* \beta^2/2$, so $-\psi_i'' \geq v_*$. Continuity of ψ_i'' extends the bound to β_c , proving $\sigma_i^2 \geq v_* > 0$. The argument uses the stable branches, not the much larger total variance at coexistence.

A.3 Uniform moments from positive contour measures

Since the minimum in (A.7) need not be twice differentiable, we first prove a Lipschitz estimate for the truncated free energies, use it to remove every cutoff in a window of width proportional to $1/L$ around β_c , and only then differentiate twice.

Lemma A.1 (Restricted contour partition derivatives). *For the fixed sufficiently large q above, there are constants $\eta, C > 0$ such that, for $i = o, d$ and all sufficiently large L ,*

$$\left| \frac{d^2}{d\beta^2} \log Z_{i,L}(\beta) \right| \leq CN, \quad |\beta - \beta_c| \leq \eta/L. \quad (\text{A.15})$$

On this interval $Z_{i,L} = Z'_{i,L}$, and

$$\left. \frac{d}{d\beta} \log[e^{d\beta N} Z_{i,L}(\beta)] \right|_{\beta=\beta_c} = -Nu_i + O(\sqrt{N}). \quad (\text{A.16})$$

Proof. We first bound the derivatives of the activities. For a general contour interior $\Lambda = \text{Int } \gamma$, use the positive matching-label sums in BCT (6.13)–(6.14). A summand indexed by an admissible family Γ has logarithm

$$F_\Gamma(\beta) = -e_o(\beta)n_o(\Gamma) - e_d(\beta)n_d(\Gamma) - \kappa(\beta)M(\Gamma) + c_\Gamma \log q, \quad (\text{A.17})$$

where the nonnegative integers n_o, n_d , with $n_o + n_d = |\Lambda \cap \Lambda_L|$, count vertices with each label, $M(\Gamma) = \sum_{\gamma' \in \Gamma} m_{\gamma'}$ counts internal contour intersections, and the integer c_Γ counts ordered components, minus one when the ordered color multiplicity is removed. These quantities and the set of admissible families do not depend on β .

To bound $M(\Gamma)$, note that BCT build contours from cubes of side $1/2$ and define contour size as the number of intersections with lattice edges (Section 4 and equation (4.5)). A contour surface is a union of faces of the half-integer cube lattice, so it meets a given lattice edge in at most a fixed number C'_d of points; compatible contours are disjoint, so the intersections counted by $M(\Gamma)$ are distinct points on distinct edges or distinct contours. A contour of Γ lies in the closure of Λ , so each edge it meets either has an endpoint in $\Lambda \cap \Lambda_L$ (at most $2d|\Lambda \cap \Lambda_L|$ edges) or crosses the bounding contour γ (at most m_γ edges, by the definition of contour size). Consequently

$$M(\Gamma) \leq C'_d(2d|\Lambda \cap \Lambda_L| + m_\gamma) \leq C_d(|\Lambda \cap \Lambda_L| + m_\gamma) =: C_d V_\gamma.$$

The bounded derivatives of e_o, e_d, κ now give $|F'_\Gamma| + |F''_\Gamma| \leq C V_\gamma$, even when the interior wraps around the torus.

For a positive sum $S = \sum_\Gamma e^{F_\Gamma}$, direct differentiation gives

$$(\log S)' = \mathbb{E}_S F'_\Gamma, \quad (\log S)'' = \mathbb{E}_S F''_\Gamma + \text{Var}_S F'_\Gamma.$$

Hence $|(\log S)'| \leq C V_\gamma$ and $|(\log S)''| \leq C(V_\gamma + V_\gamma^2)$. Applied to the numerator and denominator of BCT's activity ratios (6.18) and (6.22), with $V_\gamma = O(m_\gamma^2)$ by (A.5), these give the uniform bounds

$$|\partial_\beta \log K_i(\gamma)| \leq C m_\gamma^2, \quad |\partial_\beta^2 K_i(\gamma)| \leq C m_\gamma^4 K_i(\gamma). \quad (\text{A.18})$$

These inequalities do not require the interior phase to be stable.

For a cluster $\mathcal{C}$, write $m(\mathcal{C}) = \sum_{\gamma \in \mathcal{C}} m_\gamma$ and let $\Phi(\mathcal{C})$ denote its Ursell coefficient, including the factor for repeated polymers. The contour counting and incompatibility estimates (A.5)–(A.6) of [3] give, for some fixed $\rho > 0$,

$$\sum_{\mathcal{C}} |\Phi(\mathcal{C})| \prod_{\gamma \in \mathcal{C}} K'_i(\gamma; \beta) e^{\rho m(\mathcal{C})} \leq C N. \quad (\text{A.19})$$

The factor $e^{\rho m_\gamma}$ is absorbed by the choice of $q_0(d)$: the criterion BCT (A.5)–(A.6) holds with a slack $e^{(c\beta+1)m_\gamma}$ that grows with q , so it survives a fixed $\rho < c\beta + 1$ on the temperature interval. To sum over clusters, use the device of BCT's proof of Lemma A.2 (their Appendix A.3): every contour in Λ_L is incompatible with at least one of the N contours of size two obtained from configurations with a single disordered edge in a fixed direction, so every cluster is incompatible with one of these N fixed contours. Summing the incompatibility bound BCT (A.6) over them, with the polynomial factor absorbed in the exponential slack, proves (A.19); overcounting only increases the bound.

For the Lipschitz estimate, note that on any compact real temperature interval the minimum in (A.7) is absolutely continuous, and almost everywhere its logarithmic derivative is that of one of its two positive branches; both obey $|\partial_\beta K'_i| \leq Cm_\gamma^2 K'_i$. Termwise first differentiation is therefore justified by (A.19), since a differentiated cluster is bounded by $Cm(C)^2$ times its absolute weight (equivalently, integrate the almost-everywhere derivatives and use the same majorant to interchange sum and integral). With the ground term in (A.6), $|\partial_\beta \log Z'_{i,L}| \leq CN$ almost everywhere, so the finite-volume free energies, and hence their pointwise limits f_i^{tr} , are Lipschitz with a common constant.

Since $a_i(\beta_c) = 0$ and a_i is a difference of two C -Lipschitz functions, $a_i(\beta) \leq 2C|\beta - \beta_c|$. Choose $\eta > 0$ small enough that $2C\eta < c_0 \inf \beta$ on the fixed interval. Every torus contour has diameter at most L , so (A.8) now removes *all* cutoffs on $|\beta - \beta_c| \leq \eta/L$, where the activities are therefore the original smooth positive functions. A second derivative of a cluster product is bounded by $Cm(C)^4$ times its absolute weight, by (A.18), and the exponential factor in (A.19) absorbs this polynomial uniformly. Twice differentiating the uniformly convergent cluster expansion, and adding the ground term, proves (A.15).

For the center, set $F_{i,L}(\beta) = \log[e^{d\beta N} Z_{i,L}(\beta)]$ and take $t_N = \eta/(2\sqrt{N})$ with the sign pointing to the stable side of phase i ; $|t_N| \leq \eta/L$ since $d \geq 2$. By (A.9) and (A.10), at both β_c and $\beta_c + t_N$, $F_{i,L}(\beta) = -N\psi_i(\beta) + O(Ne^{-bL})$, and the second-derivative bound and a stable-side finite difference imply

$$F'_{i,L}(\beta_c) = -N \frac{\psi_i(\beta_c + t_N) - \psi_i(\beta_c)}{t_N} + O(N|t_N|) + O(Ne^{-bL}/|t_N|) = -Nu_i + O(\sqrt{N}).$$

The exponentially small error is negligible, proving (A.16). $\square$

A.4 Transfer to the spin energy and proof of the theorem

Lemma A.1 controls the occupied-bond count; the next lemma transfers its bounds to the spin energy exchanged with the reservoir.

Lemma A.2 (Transfer from bond fluctuations to spin energy). *In the Edwards–Sokal joint law at β_c , the conditional spin energy on each of the positive bond phase events obeys*

$$\sup_{L \geq L_0, i \in \{o, d\}} \mathbb{E} \left[\exp \left(\eta_1 \frac{|U_N - Nu_i|}{\sqrt{N}} \right) \middle| \mathcal{I}_{i,L} \right] < \infty \quad (\text{A.20})$$

for some fixed $\eta_1 > 0$ and sufficiently large fixed L_0 , where $\mathcal{I}_{o,L} = \mathcal{O}_L$ and $\mathcal{I}_{d,L} = \mathcal{D}_L$.

Proof. Let $\lambda = \log(e^\beta - 1)$ be the bond fugacity parameter, so $d\beta/d\lambda = p$, and put $B = |A|$ and $M_\sigma = -U_N$, the number of monochromatic spin bonds. Up to the constant ordered multiplicity, $e^{d\beta N} Z_{i,L}(\beta)$ is a positive sum of $e^{\lambda B} q^{k(A)}$ over the fixed phase event, so the first and second λ -derivatives of its *logarithm* $F_{i,L}$ are the conditional mean and variance of B .

Lemma A.1, the bounded derivatives of $\beta(\lambda)$, and the elementary $|F'_{i,L}| \leq CN$ give a second λ derivative bounded by CN throughout a window of width $\eta_2/\sqrt{N}$. At coexistence the first derivative is $-pNu_i + O(\sqrt{N})$, by (A.16). Taylor's theorem for log partition ratios at $\lambda_c \pm \eta_3/\sqrt{N}$ therefore proves

$$\mathbb{E} \left[e^{\eta_3 |B + pNu_i|/\sqrt{N}} \middle| \mathcal{I}_{i,L} \right] \leq C \quad (\text{A.21})$$

for a sufficiently small fixed $\eta_3 > 0$.

Conditional on the spin configuration, B is Binomial(M_σ, p), since each monochromatic bond is retained independently with probability p . For a centered Bernoulli variable, the second derivative

of its log moment-generating function is at most $1/4$; integrating twice bounds that function by $t^2/8$. Conditional independence therefore gives, for $D = B - pM_\sigma$,

$$\mathbb{E}[e^{tD} \mid \sigma] \leq e^{dNt^2/8} \quad (t \in \mathbb{R}), \quad \mathbb{E}e^{t|D|/\sqrt{N}} \leq 2e^{dt^2/8}. \quad (\text{A.22})$$

Conditioning on an event multiplies these bounds by at most its inverse probability. At coexistence all cutoffs are inactive, and (A.9) with equal free energies and the tunneling bound gives

$$P(\mathcal{O}_L) \rightarrow \frac{q}{q+1}, \quad P(\mathcal{D}_L) \rightarrow \frac{1}{q+1}, \quad P(\mathcal{T}_L) \leq Ce^{-bL^{d-1}}. \quad (\text{A.23})$$

Thus the conditional noise moments are uniformly bounded. Finally

$$|U_N - Nu_i| = |M_\sigma + Nu_i| \leq p^{-1}\{|B + pNu_i| + |D|\}.$$

Cauchy–Schwarz, (A.21), and (A.22) prove (A.20) with $2\eta_1/p \leq \eta_3$. $\square$

Proof of Theorem 4.3. For necessity, the positive midpoint phase weights, (A.13), and $\sigma_i^2 \geq v_* > 0$ satisfy Theorem 3.4, with gap $N(u_d - u_o)$ and $s_N = \sqrt{N}$.

For sufficiency, use the positive decomposition by $\mathcal{O}_L, \mathcal{D}_L, \mathcal{T}_L$ on the Edwards–Sokal space. Lemma A.2 supplies the physical-energy moment bound. The exceptional mass is at most $Ce^{-bL^{d-1}}$, while the largest reservoir amplification is $\exp[O(N^2/c_N)]$. If $c_N \gg N^{3/2}$, its logarithm is $o(\sqrt{N}) = o(L^{d/2})$. Since $d-1 \geq d/2$ for $d \geq 2$, the exceptional contribution vanishes. Theorem 3.5 therefore gives total-variation convergence on the joint spin–bond space, where the likelihood is the exact function of $U_N(\sigma)$ in (2.3). Summing over bonds gives the physical subsystem law, so projection proves the spin statement. $\square$

Optional independent quadratic momenta preserve the threshold: their centered Gamma transform adds the moment bound from Lemma 4.2, their mean shifts both centers by aN/β_c , and their variance adds a/β_c^2 to the weak phase limits. For $a > 0$, convolution also gives continuous local Gaussian energy limits; Theorem 4.3 needs no pointwise density envelope.

B Exact Gaussian models and calibration geometry

This appendix makes calibration geometry exactly solvable by reweighting Gaussian phases with a quadratic log-weight whose curvature $\kappa_N \geq 0$ stands for the entropy curvature (6.1). Gaussian reservoirs and coexistence peaks are classical [7, 8]; here they are exact models only, and their interphase tails are not substituted for the microscopic laws of Section 4. The identification $\kappa = \beta^2/(C_B/k_B)$ holds only locally, because an exactly quadratic entropy has constant curvature, not constant heat capacity.

B.1 Two scalar phases and the optimized error crossover

With φ_{s^2} the centered normal density of standard deviation $s > 0$, consider

$$p_N(E) = \frac{1}{2}\varphi_{s_N^2}(E - m_N) + \frac{1}{2}\varphi_{s_N^2}(E + m_N), \quad m_N/s_N \rightarrow \infty, \quad q_{N,t} \propto p_N e^{tE - \kappa_N E^2/2}, \quad \kappa_N \geq 0. \quad (\text{B.1})$$

The unrestricted field $t \in \mathbb{R}$ acts as the calibration; completing the square gives the transformed variance, means, and upper-phase weight

$$\tilde{s}_N^2 = \frac{s_N^2}{1 + \kappa_N s_N^2}, \quad \tilde{m}_{\pm,N} = \frac{\pm m_N + t s_N^2}{1 + \kappa_N s_N^2}, \quad r_{+,N}(t) = \frac{1}{1 + e^{-2tm_N/(1 + \kappa_N s_N^2)}}. \quad (\text{B.2})$$

Proposition B.1 (Scalar convergence and optimized crossover). *There exists a field sequence giving $d_{\text{TV}}(p_N, q_{N,t_N}) \rightarrow 0$ if and only if $\kappa_N m_N s_N \rightarrow 0$. If instead $\kappa_N m_N s_N \rightarrow a \in [0, \infty)$, then*

$$d_{\text{TV}}(p_N, q_{N,0}) \longrightarrow 2\Phi(a/2) - 1, \quad (\text{B.3})$$

$$\inf_{t \in \mathbb{R}} d_{\text{TV}}(p_N, q_{N,t}) \longrightarrow \min\{2\Phi(a/2) - 1, 1/2\}. \quad (\text{B.4})$$

Proof. Put $\rho_N = m_N/s_N$, $k_N = \kappa_N s_N^2$, and $b_N = t_N s_N$. In units of s_N , the transformed means are $(\pm\rho_N + b_N)/(1 + k_N)$, with variance $(1 + k_N)^{-1}$. If $\kappa_N m_N s_N \rightarrow 0$, then $k_N \rightarrow 0$, and $t_N = 0$ preserves the weights and makes both standardized displacements vanish, so each component converges in TV.

Necessity mirrors the label matching for Theorem B.2. For fixed $R > 0$, the windows $|E \mp m_N| \leq R s_N$ are disjoint for large N , each with target mass tending to $\frac{1}{2}[2\Phi(R) - 1]$. Since $\tilde{s}_N \leq s_N$ and $m_N/s_N \rightarrow \infty$, each transformed component has vanishing probability in one of them, so TV convergence makes each component supply one window, with weight of lower limit at least $\frac{1}{2}[2\Phi(R) - 1]$. As $R \rightarrow \infty$, the weights tend to 1/2 and the centers are $\tilde{m}_{\pm,N} = \pm m_N + O(s_N)$, in natural order. Subtracting these estimates and using the exact means gives

$$\frac{2\rho_N}{1 + k_N} = 2\rho_N + O(1), \quad \frac{k_N \rho_N}{1 + k_N} = O(1).$$

Since $\rho_N \rightarrow \infty$, $k_N \rightarrow 0$ and the variance ratio tends to one. Restricting to the matched windows and letting $R \rightarrow \infty$ gives TV convergence of each component to its target normal, so its probability above the target mean, $\Phi(\delta_{\pm,N} \sqrt{1 + k_N})$ with displacement $\delta_{\pm,N} = (b_N \mp k_N \rho_N)/(1 + k_N)$, tends to 1/2. Hence both displacements vanish; subtracting them gives $k_N \rho_N \rightarrow 0$.

For the finite crossover, $k_N \rightarrow 0$ since $k_N \rho_N \rightarrow a$ and $\rho_N \rightarrow \infty$. At $t_N = 0$ the limiting standardized components are unit-variance normals displaced by $\mp a$; separating their windows gives (B.3), using $d_{\text{TV}}(\mathcal{N}(0, 1), \mathcal{N}(a, 1)) = 2\Phi(|a|/2) - 1$.

To optimize, take any fields and a subsequence on which the transformed upper weight converges. If its limit lies in $(0, 1)$, the logistic formula in (B.2) gives $b_N = O(1/\rho_N)$, so the displacements still tend to $\mp a$; the limiting TV, a convex function of the limiting upper weight that is symmetric about 1/2 by reflection, is minimized there, giving the first term in (B.4). If one weight tends to zero, a single bounded-width Gaussian cannot cover both target windows, so the lower bound is 1/2, also when the means escape all windows. These bounds hold for arbitrary approximate minimizers. The choice $t_N = 0$ attains the first bound; for $a > 0$, $t_N = \kappa_N m_N$ centers the upper component at m_N with weight tending to one and attains 1/2; for $a = 0$ the first bound already vanishes. $\square$

With $\kappa_N = \beta^2/c_N$, $m_N = \Delta_N/2$, $s_N = \sqrt{vN}$, $\Delta_N/N \rightarrow \ell$, and $c_N/N^{3/2} \rightarrow \gamma$, one has $a = b\sqrt{v}$ with b as in Theorem 6.2, and the limits above are the balanced equal-variance cases of (6.7) and Theorem 6.4. Above $a_* = 2\Phi^{-1}(3/4) \simeq 1.34898$, near-optimal fields keep one phase accurately and discard the other.

A fixed tangent changes the required scale. Give the upper reference phase weight $w \neq 1/2$, $0 < w < 1$, and hold the tangent of the quadratic weight at the target canonical mean $\bar{E}_N = (2w - 1)m_N$, not at a self-consistent mean: $q_N \propto p_N e^{-\kappa_N (E - \bar{E}_N)^2/2}$, or $t_N = \kappa_N \bar{E}_N$. The exact phase odds are

$$\log \frac{r_{+,N}}{1 - r_{+,N}} = \log \frac{w}{1 - w} + \frac{2\kappa_N (2w - 1)m_N^2}{1 + \kappa_N s_N^2}. \quad (\text{B.5})$$

The separated windows again force $\kappa_N s_N^2 \rightarrow 0$; the weights then converge if and only if $\kappa_N m_N^2 \rightarrow 0$, which also removes the shape errors. For $m_N \asymp N$ and $s_N \asymp \sqrt{N}$, this fixed tangent needs $\kappa_N N^2 \rightarrow 0$, whereas $t_N = 0$, which like the secant calibration equalizes the reservoir weight at the two phase centers, needs only $\kappa_N N^{3/2} \rightarrow 0$.

B.2 Several exchanged quantities and phase geometry

Exact invariants limit which weights a field can restore. Let $Y \in \mathbb{R}^d$, where d counts exchanged quantities (not spatial dimensions), take distinct phase points $e_1, \dots, e_k$ with positive weights w_i summing to one, and set

$$p_N = \sum_{i=1}^k w_i \mathcal{N}(Ne_i, NI_d), \quad q_{N,t} \propto p_N \exp(t \cdot Y - \kappa_N |Y|^2/2), \quad t \in \mathbb{R}^d, \quad \kappa_N \geq 0. \quad (\text{B.6})$$

With

$$\lambda_N = (1 + \kappa_N N)^{-1}, \quad \alpha_N = \frac{\kappa_N N^2}{1 + \kappa_N N}, \quad z_N = \frac{Nt}{1 + \kappa_N N}, \quad (\text{B.7})$$

the transformed covariance is $N\lambda_N I_d$, the means are $N\lambda_N(e_i + t)$, and the weights are

$$r_{i,N} = \frac{w_i \exp(z_N \cdot e_i - \alpha_N |e_i|^2/2)}{\sum_j w_j \exp(z_N \cdot e_j - \alpha_N |e_j|^2/2)}. \quad (\text{B.8})$$

For any affine dependence $(a_1, \dots, a_k)$, meaning $\sum_i a_i = 0$ and $\sum_i a_i e_i = 0$, the field cancels, giving the exact invariant

$$\sum_i a_i \log(r_{i,N}/w_i) = -\frac{\alpha_N}{2} \sum_i a_i |e_i|^2. \quad (\text{B.9})$$

Theorem B.2 (Geometry of Gaussian reservoir convergence). *For $k \geq 2$, some field sequence in (B.6) gives TV convergence if and only if*

$$\begin{cases} \kappa_N N^{3/2} \rightarrow 0, & \text{if the phase points are cospherical,} \\ \kappa_N N^2 \rightarrow 0, & \text{otherwise.} \end{cases} \quad (\text{B.10})$$

For $k = 1$, the condition is $\kappa_N N \rightarrow 0$. Cospherical means that the points lie on a common sphere of positive radius in $\mathbb{R}^d$; in $d = 1$ a sphere is a pair of points, so three distinct scalar phase points are never cospherical.

Proof. To match component labels, note that on Y/N every transformed covariance is at most I_d/N , so if TV tends to zero, each target phase neighborhood of positive mass receives a distinct transformed component of nonvanishing weight. With k components this is a bijection, and the matched centers $\lambda_N(e_i + t_N)$ lie in bounded neighborhoods of the phase points. Along any subsequence, pass to a further one on which λ_N , $\lambda_N t_N$, and the matching permutation converge. The limiting map $e \mapsto \lambda e + a$ maps the finite phase set onto itself; its positive diameter forces $\lambda = 1$, the arithmetic centroid forces $a = 0$, and distinct phase points make the permutation the identity. Hence $\lambda_N \rightarrow 1$, $\lambda_N t_N \rightarrow 0$, and $r_{i,N} \rightarrow w_i$ with the original labels.

Isolating each matched component in expanding fluctuation windows gives its Gaussian TV convergence; in particular,

$$\sqrt{N}\{(\lambda_N - 1)e_i + \lambda_N t_N\} \longrightarrow 0 \quad \text{for every } i. \quad (\text{B.11})$$

Subtracting two equations in (B.11) for distinct phase points proves $\kappa_N N^{3/2} \rightarrow 0$.

When $\alpha_N > 0$, the phase-weight equations $r_i = w_i$ are exactly solvable if and only if

$$|e_i|^2 = 2c \cdot e_i + b \quad \text{for every } i \quad (\text{B.12})$$

for some $c \in \mathbb{R}^d$, $b \in \mathbb{R}$. Then $t_N = \kappa_N N c$ restores every weight exactly and makes the standardized mean changes $\kappa_N N^{3/2}(c - e_i)/(1 + \kappa_N N)$ vanish, proving sufficiency in the spherical case.

If (B.12) has no solution, linear algebra gives an affine dependence with $\sum_i a_i |e_i|^2 \neq 0$. Since $r_{i,N}/w_i \rightarrow 1$, (B.9) forces $\alpha_N \rightarrow 0$, equivalently $\kappa_N N^2 \rightarrow 0$, and $t_N = 0$ then proves sufficiency, including the component shapes. For a single phase, $t_N = \kappa_N N e_1$ aligns the mean exactly; matching the covariance is equivalent to $\kappa_N N \rightarrow 0$ and is necessary for any mean. $\square$

Any affinely independent set of at most $d + 1$ points is cospherical, as are some larger sets, so the number of phases alone does not fix the threshold. For scalar phase points $-1, 0, 1$, the invariant reads

$$\frac{r_{+,N} r_{-,N}}{r_{0,N}^2} = \frac{w_+ w_-}{w_0^2} e^{-\alpha_N}. \quad (\text{B.13})$$

As for three support points in Theorem 3.1, the threshold is $\kappa_N N^2 \rightarrow 0$.

B.3 The exact optimum when some phases must disappear

For non-cospherical points with $\kappa_N N^2 \rightarrow \infty$ some phase weight is lost; the next theorem gives the optimal loss in the regime $\kappa_N N^{3/2} \rightarrow 0$, where component shapes are still preserved. An empty-sphere contact set is the set of phase indices nearest to a center c (its points lie on a sphere about c with no phase point inside),

$$F(c) = \arg \min_{1 \leq i \leq k} |e_i - c|^2, \quad c \in \mathbb{R}^d. \quad (\text{B.14})$$

These are the contact sets of the lower-paraboloid, or Delaunay, construction [18]. Concentration of exponential families on faces has a general theory [15]; the theorem here determines the exact optimized TV loss under the specified diverging quadratic parameter. As only finitely many subsets occur, $W_* := \max_c \sum_{i \in F(c)} w_i$ is attained.

Theorem B.3 (Optimal Gaussian phase loss). *For the model (B.6), if*

$$\kappa_N N^{3/2} \rightarrow 0, \quad \kappa_N N^2 \rightarrow \infty, \quad (\text{B.15})$$

then

$$\lim_{N \rightarrow \infty} \inf_{t \in \mathbb{R}^d} d_{\text{TV}}(p_N, q_{N,t}) = 1 - W_*. \quad (\text{B.16})$$

The optimization includes field sequences of unbounded norm.

Proof. For the upper bound, fix c and take $t_N = \kappa_N N c$. Up to a common constant, the phase score in (B.8) is $-\alpha_N |e_i - c|^2/2$, so as $\alpha_N \rightarrow \infty$ the weights converge to the reference weights conditioned on $F(c)$. With shape changes vanishing by the first condition in (B.15), the limiting TV is $1 - \sum_{F(c)} w_i$; maximize over c .

For the lower bound, take any fields. On a subsequence with $|t_N| \geq \varepsilon > 0$, put $u_N = t_N/|t_N|$ and $H_N = \max_i u_N \cdot e_i$. Indices with $u_N \cdot e_i < H_N - N^{-1/4}$ carry vanishing weight: relative to a maximizer, their linear score is lower by at least $c_1 N^{3/4}$ for a fixed $c_1 > 0$, while their quadratic score differs only by $O(\alpha_N) = o(\sqrt{N})$. For the remaining indices the projected transformed mean per particle is at least

$$\lambda_N(H_N - N^{-1/4} + |t_N|) \geq H_N + \varepsilon/2$$

for all large N , uniformly also when $|t_N| \rightarrow \infty$. The target mixture has negligible mass where $u_N \cdot Y/N > H_N + \varepsilon/4$, whereas the transformed law has mass tending to one there, so TV tends to one.

It remains to treat $t_N \rightarrow 0$. On a subsequence where all weights converge, let I be the nonempty set of indices with positive limiting weight. As $\lambda_N \rightarrow 1$, the transformed means divided by N approach their own e_i , and fixed small disjoint neighborhoods of the phase points give

$$\liminf d_{\text{TV}}(p_N, q_{N,t_N}) \geq 1 - \sum_{i \in I} w_i. \quad (\text{B.17})$$

To constrain I , fix $i_0 \in I$ and put $\hat{c}_N = z_N/\alpha_N$; the exact score ratios imply

$$\begin{aligned} \hat{c}_N \cdot (e_i - e_{i_0}) - \frac{1}{2}(|e_i|^2 - |e_{i_0}|^2) &\rightarrow 0 & (i \in I), \\ \hat{c}_N \cdot (e_j - e_{i_0}) - \frac{1}{2}(|e_j|^2 - |e_{i_0}|^2) &\leq o(1) & (1 \leq j \leq k). \end{aligned}$$

For the inequalities, $r_{i_0,N}$ is bounded below and $r_{j,N} \leq 1$, so the log-ratio bound divided by α_N vanishes. A fixed finite linear system satisfied up to vanishing errors cannot be infeasible: Farkas' alternative would supply fixed multipliers annihilating the unknown vector and yielding a strictly negative constant, contradicting the vanishing errors. So some finite c satisfies the limiting system exactly, even if $\hat{c}_N$ has no bounded subsequence. Then $F(c) \supseteq I$, so $\sum_{i \in I} w_i \leq W_*$, and (B.17) gives the lower bound along every subsequence of approximate minimizers. $\square$

For three scalar points $-1, 0, 1$, the largest retained sets in Theorem B.3 are the two adjacent pairs, so the limiting optimal error is $\min(w_-, w_+)$, or $1/3$ for equal weights.

B.4 Anisotropic curvature: the effective metric

Keep the common covariance $N I_d$ but replace $\kappa_N I_d$ by a symmetric positive-semidefinite curvature matrix B_N . Completing squares gives

$$C_N = N(I_d + N B_N)^{-1}, \quad M_N = B_N(I_d + N B_N)^{-1}, \quad z_N = N(I_d + N B_N)^{-1} t_N, \quad (\text{B.18})$$

as the transformed covariance, effective metric, and effective field. The means are $N(I_d + N B_N)^{-1}(e_i + t_N)$, and $r_i \propto w_i \exp[z_N \cdot e_i - N^2 e_i^T M_N e_i / 2]$. The affine invariants and exact weight-restoration equations therefore use M_N , not B_N . The sphere becomes an ellipsoid, or a point at zero radius, for positive-definite M_N . For singular $M_N \neq 0$ it is an elliptic cylinder, or an affine subspace at zero radius, if the linear coefficient lies in the range, and a paraboloid, possibly with flat directions, if the coefficient has a nonzero nullspace component. For $M_N = 0$ it is a hyperplane if the coefficient is nonzero, or the whole space if the coefficient and compatible constant vanish. Inconsistent level constants give an empty locus in the completed-square or constant cases. These finite- N algebraic statements do not assert the exponent classification (B.10) for arbitrary anisotropic sequences or unequal phase covariance matrices.

C Interfacial tails and accuracy diagnostics

This appendix gives the capillarity model behind the large- γ range (6.17) of the two-dimensional boundary law and the assumption $\alpha > 1/2$ in the density-envelope route to the boundary law ($\alpha = 1/2$ is the two-dimensional interfacial exponent). It then contrasts full-state TV with histogram norms and barrier accuracy.

C.1 A capillarity model for the two-dimensional boundary

Biskup, Chayes, and Kotecký established the competition between background fluctuations and droplet surface costs [1]. Leung and Zia located the droplet-to-strip transition on a periodic square at fixed excess volume fraction [26], Neuhaus and Hager analyzed it for first-order simulations [28], and Kim, Keyes, and Straub exhibited droplet and strip energy regions in a toroidal two-dimensional Potts system [22]. These results motivate the model below; none supplies our assumed full density envelope.

Assume $Y_N = (E - e_{-,N})/\Delta_N$ is supported in a fixed compact interval K containing $[0, 1]$ and satisfies a large-deviation principle with speed $\sqrt{N}$ and good rate I ; the model is stated for $d = 2$, where this speed is both the surface scale $N^{(d-1)/d}$ and the fluctuation scale. Assume also that I is continuous and finite on $[0, 1]$, infinite on $K \setminus [0, 1]$, zero exactly at $0, 1$, and positive on $(0, 1)$. Compact support is a simplification, not a claim about systems with unbounded kinetic energy.

Proposition C.1 (Capillarity variational law). *For $\Delta_N/N \rightarrow \ell > 0$ and $c_N/N^{3/2} \rightarrow \gamma > 0$, reweighting by the exact reservoir (2.2) with the secant calibration (2.10) gives a large-deviation principle at speed $\sqrt{N}$ with rate*

$$I_\gamma(\theta) = I(\theta) - J_\gamma(\theta) - \min_{0 \leq u \leq 1} \{I(u) - J_\gamma(u)\}, \quad J_\gamma(\theta) = \frac{\beta^2 \ell^2}{2\gamma} \theta(1 - \theta). \quad (\text{C.1})$$

If

$$\gamma_* = \frac{\beta^2 \ell^2}{2} \sup_{0 < \theta < 1} \frac{\theta(1 - \theta)}{I(\theta)} < \infty, \quad (\text{C.2})$$

then $\gamma > \gamma_*$ leaves only the endpoints as rate minimizers. For $\gamma < \gamma_*$ all minimizing energies lie away from the endpoints and $d_{\text{TV}}(P_N, Q_N) \rightarrow 1$.

Proof. Equation (6.14), expanded on K , gives $h_N/\sqrt{N} \rightarrow J_\gamma$ uniformly; the cutoff lies beyond K for large N because $c_N/\Delta_N \rightarrow \infty$. For this continuous bounded tilt, the large-deviation upper bound on a finite cover by sets where the tilt varies by at most ϵ , with the lower bound on an open neighborhood of each point, gives

$$\lim_N \frac{1}{\sqrt{N}} \log \mathbb{E} e^{h_N} = \max_{\theta \in K} \{J_\gamma(\theta) - I(\theta)\}.$$

The same bounds for restricted integrals prove (C.1). If $\gamma > \gamma_*$, $I(\theta) > J_\gamma(\theta)$ throughout the interior. If $\gamma < \gamma_*$, the minimum of $I - J_\gamma$ is strictly negative, whereas its endpoint values are zero, so by compactness and continuity its minimizing set lies a positive distance from the endpoints. A neighborhood of that set has reservoir probability tending to one and canonical probability tending to zero, so $d_{\text{TV}}(P_N, Q_N) \rightarrow 1$. $\square$

For $\gamma > \gamma_*$ the proposition gives macroscopic energy concentration, not full-state TV or the fluctuation law within a phase. At $\gamma = \gamma_*$ the model does not identify the weights of tied minimizers: multiplying the law near them by positive constants, or by factors with logarithm $o(\sqrt{N})$, leaves the rate unchanged but can change those weights or select one minimizer.

For an explicit square-torus model, set

$$I(\theta) = \tau \min\{2\sqrt{\pi\theta}, 2, 2\sqrt{\pi(1-\theta)}\}, \quad 0 \leq \theta \leq 1, \quad \tau > 0. \quad (\text{C.3})$$

The branches are an isotropic circular droplet, a slab with two interfaces, and the complementary bubble: the isotropic-tension form of the standard square-torus construction [26, 28], with the droplet-to-strip crossover at $\theta = 1/\pi$. Equation (C.3) is a model input, not a Wulff law derived for the lattice Potts model.

Corollary C.2 (Square-torus minimizers). *For (C.3), $\gamma_* = \beta^2 \ell^2 / (16\tau)$. Above γ_* the only minimizers are 0, 1; below it the unique minimizer is 1/2; at equality the minimizers are 0, 1/2, 1.*

Proof. For all θ ,

$$I(\theta) \geq 8\tau\theta(1 - \theta), \quad (\text{C.4})$$

with equality exactly at 0, 1/2, 1. The slab branch obeys this because $\theta(1 - \theta) \leq 1/4$. For the droplet branch, away from $\theta = 0$ the bound follows from $8\sqrt{\theta}(1 - \theta) \leq 16/(3\sqrt{3}) < 2\sqrt{\pi}$; the bubble branch is symmetric. Equality at 1/2 gives the stated γ_* . Put $A = \beta^2 \ell^2 / (2\gamma)$. If $A < 8\tau$, only endpoints minimize $I - A\theta(1 - \theta)$. If $A = 8\tau$, equality in (C.4) identifies the three minimizers. If $A > 8\tau$, using $I(1/2) = 2\tau$ gives

$$[I(\theta) - A\theta(1 - \theta)] - [I(1/2) - A/4] \geq (A - 8\tau)(\theta - 1/2)^2,$$

which is strictly positive off 1/2. □

C.2 Histogram norms can hide a persistent error

An unscaled density norm can vanish while TV stays fixed, even within one phase. Let p_σ be the density of $\mathcal{N}(0, \sigma^2)$ and q_σ that of $\mathcal{N}(0, r\sigma^2)$, for fixed $0 < r < 1$. Gaussian integration gives

$$\|p_\sigma - q_\sigma\|_2^2 = \frac{1}{2\sqrt{\pi}\sigma} \left[1 + r^{-1/2} - 2\sqrt{\frac{2}{1+r}} \right] \rightarrow 0 \quad (\sigma \rightarrow \infty), \quad (\text{C.5})$$

although

$$d_{\text{TV}}(p_\sigma, q_\sigma) = 2 \left[\Phi\left(\sqrt{\frac{\log r}{r-1}}\right) - \Phi\left(\sqrt{\frac{r \log r}{r-1}}\right) \right] > 0. \quad (\text{C.6})$$

The densities cross at $|E|/\sigma = \sqrt{r \log r / (r - 1)}$, and integrating their difference between the crossings gives (C.6). For histogram probabilities in bins of fixed width $a > 0$, a rescaled Riemann sum gives $\sum_j (P_j - Q_j)^2 \sim a \|p_\sigma - q_\sigma\|_2^2$, so the Euclidean histogram norm decays as $\sigma^{-1/2}$, or $N^{-1/4}$ when $\sigma \asymp \sqrt{N}$, although the variance ratio and the TV error stay fixed.

This is the distinction promised in Section 1: the finite-bath Potts comparison of Griffin, Matty, and Swendsen measures agreement by the unweighted Euclidean norm of energy-probability differences, their equation (23), and reports its decay with size [20]; the example shows that such decay is compatible with a fixed full-state error.

C.3 Barrier accuracy requires a separate scale

Let the canonical and reservoir energy densities p and q be positive at specified energies e_- and $E^\dagger$, with fixed-energy log barriers $B_P = \log[p(e_-)/p(E^\dagger)]$ and $B_Q = \log[q(e_-)/q(E^\dagger)]$. With $h(e_-) = 0$, normalization cancels exactly:

$$B_Q - B_P = -h(E^\dagger). \quad (\text{C.7})$$

The same holds for ratios of point probabilities in a discrete energy law. For a quadratic weight balanced at the endpoints with gap $\Delta = 2m$, the midpoint gain is $\kappa m^2/2 = \kappa \Delta^2/8$; for the exact secant-calibrated reservoir,

$$h\left(\frac{e_- + e_+}{2}\right) = c \log \cosh\left(\frac{\beta \Delta}{2c}\right), \quad (\text{C.8})$$

which follows from (6.14) at $\theta = 1/2$ and is positive.

If $c \gg \Delta$, the barrier lowering is $\beta^2 \Delta^2 / (8c) [1 + O((\Delta/c)^2)]$. For $\Delta \rightarrow \infty$ and inverse temperature bounded away from zero and infinity, the lowering vanishes if and only if $c \gg \Delta^2$, because $\log \cosh x$ lies between a positive constant times $\min(x^2, |x|)$ and $x^2/2$; the lower bound follows from continuity of the corresponding ratios on $0 < |x| \leq 1$ and $|x| \geq 1$, with positive limits at zero and infinity. Along a subsequence with Δ/c bounded below the barrier error is thus of order at least Δ , while in the other regime it is comparable to Δ^2/c .

For an extensive gap, absolute accuracy of the midpoint barrier therefore requires $c \gg N^2$, whereas the two-phase full-state TV error can vanish for $c \gg N^{3/2}$. Relative accuracy is less demanding: for a canonical barrier $B_P \asymp N^{(d-1)/d}$ and $c \gg N$, it requires $c \gg N^2/B_P \asymp N^{1+1/d}$, at most $N^{3/2}$ for $d \geq 2$. These statements concern density ratios at specified energies, not shifted peaks, saddle points, or dynamical transition rates.

D Shared-reservoir extensions: fixed phase counts and linear capacity

This appendix extends Section 5 under the same hypotheses (5.1)–(5.2).

D.1 Selecting a fixed phase count among finitely many copies

For a fixed integer $m \geq 2$, take the reference law $\Pi_N^{(m)} = P_N^{\otimes m}$, let K_N^+ count the plus-phase copies, and fix $k \in \{0, \dots, m\}$. Just as the calibration (5.3) selects opposite phases for two copies, centering the reservoir at the energy of k plus-phase and $m - k$ minus-phase copies,

$$M_{m,k,N} = (m - k)e_{-,N} + ke_{+,N}, \quad \mathcal{E}_N = M_{m,k,N} + c_N/\beta_N, \quad (\text{D.1})$$

selects that phase count. Denote the reweighted m -copy law by $Q_N^{[m]}$ and its one-copy marginals by $Q_{j,N}^{[m]}$. Write $f = k/m$ and

$$p_k = \binom{m}{k} w_+^k w_-^{m-k}, \quad H(f) = -f \log f - (1 - f) \log(1 - f), \quad (\text{D.2})$$

with the usual zero-term convention at $f = 0, 1$. The total correlation $D_{\text{KL}}(Q_N^{[m]} \| \bigotimes_j Q_{j,N}^{[m]})$ measures the dependence among the copies; for $m = 2$ it is the mutual information.

Proposition D.1 (Fixed-count selection and total correlation). *Under the hypotheses of Theorem 5.1,*

$$d_{\text{TV}}(Q_N^{[m]}, \Pi_N^{(m)}(\cdot \mid K_N^+ = k)) \rightarrow 0, \quad (\text{D.3})$$

$$d_{\text{TV}}(Q_N^{[m]}, \Pi_N^{(m)}) \rightarrow 1 - p_k, \quad (\text{D.4})$$

$$d_{\text{TV}}(Q_{j,N}^{[m]}, P_N) \rightarrow |f - w_+|, \quad (\text{D.5})$$

$$D_{\text{KL}} \left(Q_N^{[m]} \left\| \bigotimes_{j=1}^m Q_{j,N}^{[m]} \right. \right) \rightarrow mH(f) - \log \binom{m}{k}. \quad (\text{D.6})$$

Proof. On $K_N^+ = k$, the energy deviation from $M_{m,k,N}$ is tight on scale $\sqrt{N}$ because m is fixed. On any other count it equals $(K_N^+ - k)\Delta_N + O_{\Pi_N^{(m)}}(\sqrt{N})$ conditionally. The proof of Theorem 5.1 therefore gives L^1 convergence of the bounded weight to $\mathbf{1}_{\{K_N^+ = k\}}$, proving (D.3). All phase assignments with

count k have the same canonical probability, so conditioning selects them uniformly. Each marginal has plus-phase probability f and the unchanged phase-conditional microscopic laws, which proves the two total-variation formulas.

The bounded-likelihood argument from Proposition 5.2 gives $D_{\text{KL}}(Q_N^{[m]} \parallel \Pi_N^{(m)}) \rightarrow -\log p_k$ and

$$D_{\text{KL}}(Q_{j,N}^{[m]} \parallel P_N) \rightarrow f \log \frac{f}{w_+} + (1-f) \log \frac{1-f}{w_-}.$$

Subtracting the m marginal divergences by the relative-entropy chain rule gives (D.6). The argument also covers $k = 0, m$, since $r \log r$ is continuous at zero. $\square$

When $w_+ = k/m$ with $0 < k < m$, every full-state one-copy marginal becomes canonical and the limiting total correlation is $-\log p_k$ (one bit for $m = 2$). No claim is made for m growing with N : the phase probability and normalization bounds in the proof are not uniform in m .

D.2 The centered reservoir at linear capacity

At capacity proportional to N , the reservoir also correlates the fluctuations within an opposite-phase assignment. Retain the centered calibration (5.3), and strengthen the phase assumption to

$$\frac{E_N - e_{i,N}}{\sqrt{N}} \mid A_{i,N} \Rightarrow \mathcal{N}(0, v_i), \quad v_- > 0, \quad v_+ > 0, \quad c_N/N \rightarrow \gamma \in (0, \infty). \quad (\text{D.7})$$

Define

$$\alpha = \beta^2/\gamma, \quad V = v_- + v_+, \quad A_\gamma = (1 + \alpha V)^{-1/2}, \quad p_O = 2w_-w_+. \quad (\text{D.8})$$

These results concern only this calibration; no optimality over other calibrations is claimed.

Theorem D.2 (Gaussian fluctuation tilt at linear capacity). *Under (5.1)–(5.2) and (D.7), the two opposite-phase assignments each have limiting probability 1/2 under Q_N , and the same-phase assignments disappear. Within an opposite assignment, order the standardized energies by phase. Their limiting law is centered Gaussian with covariance*

$$\Sigma_\gamma = D - \frac{\alpha}{1 + \alpha V} v v^\top, \quad D = \text{diag}(v_-, v_+), \quad v = (v_-, v_+)^\top. \quad (\text{D.9})$$

The one-copy variances and squared correlation coefficient of this limiting Gaussian law are

$$\hat{v}_i = \frac{v_i(1 + \alpha v_j)}{1 + \alpha V} \quad (j \neq i), \quad \rho_\gamma^2 = \frac{\alpha^2 v_- v_+}{(1 + \alpha v_-)(1 + \alpha v_+)}. \quad (\text{D.10})$$

In particular $0 < \hat{v}_i < v_i$, and the correlation is negative.

Proof. For a state in phase i write $Y_{i,N} = (E_N - e_{i,N})/\sqrt{N}$. On an opposite assignment, the exact weight (5.4) converges uniformly on compact standardized windows to

$$W_\gamma(y_-, y_+) = \exp[-\alpha(y_- + y_+)^2/2]. \quad (\text{D.11})$$

On a same-phase assignment its argument $x_N = \beta_N T_N / c_N$ tends to the nonzero constant $\pm \beta \ell / \gamma$. If that constant is negative, the strict inequality $x + \log(1 - x) < 0$ makes the log weight at most a negative constant times N on each compact window. For a positive constant, the cutoff gives zero whenever $x_N \geq 1$, and otherwise $x_N + \log(1 - x_N) \leq -x_N^2/2$, which also gives suppression, even when the limiting center meets the cutoff.

Conditional weak convergence, tightness, and $0 \leq W_N \leq 1$ imply

$$z_N \longrightarrow p_O A_\gamma, \quad A_\gamma = \int W_\gamma(y_-, y_+) \varphi_{v_-}(y_-) \varphi_{v_+}(y_+) dy_- dy_+. \quad (\text{D.12})$$

The integral equals $(1 + \alpha V)^{-1/2}$ because the canonical sum has variance V . Exchange symmetry makes the two surviving assignment weights equal. Multiplication by (D.11) adds $\alpha \mathbf{1}\mathbf{1}^\top$ to the Gaussian precision matrix D^{-1} . Inverting it gives (D.9); its entries give (D.10). $\square$

The covariance statement concerns weak convergence of standardized energies, not total-variation convergence of microscopic phase laws to Gaussian laws. A full-state approximation is nevertheless available: keep the microscopic reference Π_N , replace W_N by $\mathbf{1}_{O_N} W_\gamma(Y_{S_1, N}, Y_{S_2, N})$, and normalize; the result approaches Q_N in full-state TV because the two bounded unnormalized weights approach each other in $L^1(\Pi_N)$.

Proposition D.3 (Full marginal error at linear capacity). *Under the same assumptions, either full-state one-copy marginal has the limit*

$$\lim_{N \rightarrow \infty} d_{\text{TV}}(Q_N^{(j)}, P_N) = \frac{1}{2} \sum_{i \in \{-, +\}} \int_{\mathbb{R}} \left| \frac{1}{2} \varphi_{\hat{v}_i}(y) - w_i \varphi_{v_i}(y) \right| dy. \quad (\text{D.13})$$

For balanced weights this is strictly positive and equals

$$\sum_{i \in \{-, +\}} \left[\Phi\left(\frac{x_i}{\sqrt{\hat{v}_i}}\right) - \Phi\left(\frac{x_i}{\sqrt{v_i}}\right) \right], \quad x_i^2 = \frac{\hat{v}_i v_i \log(v_i / \hat{v}_i)}{v_i - \hat{v}_i}, \quad (\text{D.14})$$

where Φ is the standard normal CDF.

Proof. On a phase- i window, integrate the kernel against the other copy's energy law. Its opposite-phase contribution converges, uniformly on compact y -intervals, to $w_j g_i(y)$, where

$$g_i(y) = \frac{1}{\sqrt{1 + \alpha v_j}} \exp\left[-\frac{\alpha y^2}{2(1 + \alpha v_j)}\right], \quad \frac{g_i(y)}{A_\gamma} = \frac{\varphi_{\hat{v}_i}(y)}{\varphi_{v_i}(y)}. \quad (\text{D.15})$$

For uniformity, restrict the other standardized energy to a fixed compact interval, where the kernel convergence is uniform and its continuous limit is uniformly continuous. Weak convergence then gives uniformity in y by a finite grid argument. Tightness and the bound by one control the discarded other-copy tail. The same-phase contribution vanishes by the same argument and the cutoff estimate in Theorem D.2.

The exact marginal likelihood therefore tends on compact phase windows to $g_i(y)/(2w_i A_\gamma)$ and is globally bounded by $1/z_N$, so tightness controls the rest of the marginal TV integral, and weak Gaussian convergence gives

$$\frac{1}{2} \sum_i w_i \int \left| \frac{g_i(y)}{2w_i A_\gamma} - 1 \right| \varphi_{v_i}(y) dy,$$

which is (D.13). At balance it is one half the sum of the two Gaussian total-variation distances. The two centered Gaussian densities cross at $\pm x_i$; integration between the crossings gives (D.14). $\square$

The same bounded-likelihood argument gives the joint error

$$\lim d_{\text{TV}}(Q_N, \Pi_N) = \frac{1 - p_O}{2} + \frac{p_O}{2} \mathbb{E} \left| \frac{e^{-\alpha G^2/2}}{p_O A_\gamma} - 1 \right|, \quad G \sim \mathcal{N}(0, V). \quad (\text{D.16})$$

Equivalently, if $t_\gamma^2 = -2 \log(p_O A_\gamma)/\alpha$, which is positive because $p_O \leq 1/2$ and $A_\gamma < 1$, this limit is

$$2\Phi\left(t_\gamma \sqrt{\frac{1+\alpha V}{V}}\right) - 1 - p_O \left[2\Phi\left(\frac{t_\gamma}{\sqrt{V}}\right) - 1\right].$$

This follows by integrating over the set $|G| < t_\gamma$ where the likelihood exceeds one, using variance $V/(1+\alpha V)$ under the tilted law.

Proposition D.4 (Phase and fluctuation information at linear capacity). *Under the hypotheses of Theorem D.2,*

$$\lim_{N \rightarrow \infty} I_{Q_N}(X_1; X_2) = \log 2 - \frac{1}{2} \log(1 - \rho_\gamma^2) = \log 2 + \frac{1}{2} \log \frac{(1 + \alpha v_-)(1 + \alpha v_+)}{1 + \alpha V}. \quad (\text{D.17})$$

The second term is strictly positive and is independent of w_-, w_+ .

Proof. Boundedness and continuity of $x \log x$ on $[0, 1]$ allow the same compact-window argument as for (D.12). Under the Gaussian tilt the sum of standardized energies has variance $V/(1+\alpha V)$. Therefore

$$\lim D_{\text{KL}}(Q_N \| \Pi_N) = -\log p_O + \frac{1}{2} \log(1 + \alpha V) - \frac{\alpha V}{2(1 + \alpha V)}. \quad (\text{D.18})$$

For the marginals, the compact convergence of their likelihoods in Proposition D.3, their uniform bound by $1/z_N$, and uniform continuity of $r \log r$ give

$$\lim D_{\text{KL}}(Q_N^{(j)} \| P_N) = \frac{1}{2} \sum_i \log \frac{1}{2w_i} + \frac{1}{2} \sum_i D_{\text{KL}}(\mathcal{N}(0, \hat{v}_i) \| \mathcal{N}(0, v_i)). \quad (\text{D.19})$$

These limits concern bounded microscopic likelihood integrals, not convergence of differential entropies. The Gaussian divergence is $\frac{1}{2}[\hat{v}_i/v_i - 1 - \log(\hat{v}_i/v_i)]$. Substituting it and (D.18) into the finite-divergence chain rule (5.12) gives (D.17). $\square$

As $\gamma \rightarrow \infty$, the linear-capacity formulas approach the limits (5.8) and (5.9) for the window (5.5). This compares limiting formulas only; Theorem 5.1 treats that window directly.

The positive-variance hypotheses hold for the plain short-range Potts models by (A.13)–(A.14), also after the balancing shift, and for the mean-field model when $a > 0$. The pure-spin mean-field model has $v_+ = 0$, so it is outside the linear-capacity statements but remains covered by the opposite-phase and fixed-count selection results.

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